Private Preferences for Jobs and Labor Market Dynamics

Cedomir Malgieri[†] Martin Souchier [‡]

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Abstract

We build a new dynamic search model in which workers have heterogeneous preferences for jobs that are not observed by employers, giving rise to monopsony. Employers make wage offers based on beliefs about workers' preferences. These beliefs evolve endogenously over time because workers' mobility decisions reveal information about how much they like their jobs. We estimate the model using matched employer-employee data from France and quantify how preferences for jobs shape wage dynamics, markdowns, and worker mobility. Markdowns are heterogeneous both across and within firms, generating inefficient labor market flows. Quantitatively, at least one third of job-to-job transitions are inefficient: workers sometimes move to lower-surplus jobs because those jobs let workers enjoy higher rents. Policies that limit the information available to prospective employers, such as salary history bans, increase wages and reduce inequality, but they also reduce allocative efficiency.

Keywords: Search Frictions, Monopsony, Preferences, Wage Dynamics, Sorting

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[†]W.P. Carey School of Business, Arizona State University. Email: cmalgie1@asu.edu

[‡]The Wharton School, University of Pennsylvania. Email: souchier@wharton.upenn.edu

1 Introduction

There is growing evidence from survey data that workers have large and heterogeneous preferences for jobs (Mas and Pallais, 2017; Maestas et al., 2023; Caldwell, Haegele, and Heining, 2025) and this heterogeneity in preferences has been considered a key source of monopsony power for firms (Card et al., 2018; Card, 2022; Kline, 2025). How do these preferences influence wage dynamics and worker mobility across jobs? These issues are closely connected: workers' mobility decisions depend on the wage offers they receive from firms, and the wages that firms offer depend on the labor supply they face and thus on workers' mobility decisions.

Existing models cannot be used to answer this question. The typical monopsony models used to study wages are static (Berger, Herkenhoff, and Mongey, 2022; Lamadon, Mogstad, and Setzler, 2022) so they cannot be used to study how preferences influence wage dynamics and employer-to-employer mobility. Moreover, labor relationships are long-lasting and employers can learn about worker preferences over time (Coase, 1972). By contrast, the models used in the search literature are dynamic, but they assume that preferences are observable to firms (Berger et al., 2023; Lamadon et al., 2024) so they are not a source of monopsony as in Card (2022). These models are well suited to study amenities that are valued similarly by all workers but much less to study preferences that are heterogeneous across workers. In this sense, there does not exist a model in which heterogeneity in worker preferences leads to monopsony and influences labor market dynamics.

In this paper, we build and estimate a new dynamic search model in which preference heterogeneity leads to monopsony and use it to evaluate how these preferences influence labor market dynamics. In this model, firms make wage offers given the beliefs they have about worker preferences. These beliefs change over time as firms observe the worker's decision to stay or switch jobs, which reveals information about their preferences. We estimate the size and heterogeneity in preferences using information on employer-to-employer (EE) transitions from French administrative data. The model replicates several untargeted moments on how preferences affect wages as well as estimates of preferences from survey data. Markdowns are heterogeneous both between and within firms, and they account for a substantial share of wage inequality. Both the level and the hetorogeneity in markdowns generate inefficient labor market flows, from unemployment to employment and across jobs. Quantitatively, at least one third of job-to-job transitions are inefficient, as workers often move from high-surplus jobs to low-surplus jobs to benefit from higher rents. We use our model to study the effects of policies such as salary history bans, that limit the information available to prospective employers. These policies increase wages and reduce wage inequality, but they also reduce

allocative efficiency.

We first provide new evidence that preferences drive worker mobility across jobs and that these preferences are heterogeneous across workers. First, we document that 38% of workers switching jobs experience wage cuts. We show that these transitions cannot be fully explained by layoffs, godfather shocks, or expectations of higher wage growth at the next job. We instead provide suggestive evidence using distance to work that job characteristics improve when workers switch to jobs with lower pay. Second, we show that 31% of job-to-job transitions are toward lower-ranked firms. To do so, we rank firms using the revealed-preference algorithm of Sorkin (2018). Since this ranking captures the component of job value that is common across workers, moves toward lower-ranked firms point to idiosyncratic factors as important drivers of mobility. We show that these transitions to lower-ranked firms cannot be accounted for by layoffs, godfather shocks, match productivity or heterogeneity across groups (e.g. age) that is observable to firms and thus could be priced into wages. Although our approach relies on labor market flows, these results are consistent with recent survey evidence that finds that these preferences are large and heterogeneous across workers.

To account for this evidence, we build a new dynamic search model in which worker preferences for jobs lead to monopsony. The model has random search, job-to-job transitions and counteroffers in the spirit of Postel-Vinay and Robin (2002) and Cahuc, Postel-Vinay, and Robin (2006). Firms are heterogeneous in terms of productivity and amenities but both are publicly observable so they are priced into wages. By contrast, we introduce into this model preferences that are private to workers in the sense that they are heterogeneous across workers and not observed by firms. This idea dates back at least to Robinson (1933) and is the workhorse assumption in the static models used in the monopsony literature. In our dynamic model, firms make wage offers based on beliefs about workers' preferences and these beliefs are endogenously updated over time as firms observe workers' mobility decisions. This learning process introduces a new state variable, beliefs about worker preferences, and gives rise to a fixed point problem in which wage offers depend on beliefs, beliefs depend on mobility decisions, and mobility decisions depend on wage offers.

To fix ideas, consider a worker who recently joined a new firm. She may place a high value on specific attributes of the job, such as her colleagues, the location, or the nature of the work, but her employer does not observe these preferences. Over time, however, the employer can infer the worker's preference for the job by observing her mobility decisions. If the worker receives outside offers from other firms and repeatedly chooses to stay, the employer will rationally conclude that she values the job highly. Then, this worker will be less likely to receive a wage increase the next time she meets with a potential new employer. This illustrative example suggests that information about how much a given worker likes her

job interacts in a fundamental way with job-to-job mobility and wages.

Consider how wages are set for unemployed workers in this model. Firms make take-it-or-leave-it wage offers. As in static monopsony, the probability of attracting a worker is increasing in the wage. In our model, monopsony also has a dynamic component: firms recognize that the wage they offer today reveals valuable information about the worker's preferences. If a firm offers a low wage and the worker accepts, the firm infers that the worker has a strong preference for the job. This information is valuable in the future when the same worker will receive an outside offer: a worker who strongly values her current job is less likely to leave and less likely to require a wage increase to be retained. Workers understand this trade-off. As a result, a worker may eventually leave a job that she likes to avoid revealing her attachment to the firm and weakening her future bargaining position. Moreover, our model also provides a theory for why firms may *choose* not to respond to outside offers and instead lose workers to competitors, even when raising wages was feasible and would have allowed them to retain those workers.

We estimate the model using matched employer-employee data from France. The estimation targets key moments on labor market flows and on changes in productivity, wages, and firm values during job-to-job transitions. We then assess the model's fit using untargeted moments on how heterogeneity in worker preferences for jobs affect wages and mobility. The model implies an average wage markdown close to 20%, consistent with Berger, Herkenhoff, and Mongey (2022). It also implies an acceptance rate of job offers from unemployment of about 50%, in line with Faberman et al. (2022). We also validate our model on evidence from survey data on the size of these preferences. Specifically, Caldwell, Haegele, and Heining (2025) finds that workers require a 7-18% wage increase to move to an otherwise identical firm.[1] Our model is consistent with this evidence, and the premium arises because workers sort into jobs they like. Finally, the model predicts that the share of job-to-job transitions toward lower-ranked firms is higher at low tenure, a pattern we also document in the French matched employer-employee data. In the model, this pattern emerges endogenously because workers with strong preferences for their current jobs are more likely to leave at higher tenure to avoid revealing their attachment and weakening their bargaining position.

Wage markdowns are heterogeneous both across and within firms. Quantitatively, about 60% of markdown dispersion is explained by variation within firm types, similar to Volpe (2024). Within firm types, markdowns vary because workers have different labor market histories. This heterogeneity is important for wage inequality: decomposing wage dispersion

[1]These estimates are obtained after controlling for firm-specific non-wage amenities in several ways. For example, the "otherwise identical firm" is specified as a particular firm, and the results are robust to whether the firm is described as identical by the worker or by the researcher. In additional robustness exercises, workers are asked to consider a firm with the same benefits and the same distance from their residence.

into marginal products and markdowns shows that heterogeneity in markdowns accounts for 57% of wage dispersion. However, wage inequality does not fully capture inequality in flow utility, and the gap is mostly due to idiosyncratic preferences. While firm-specific non-wage amenities contribute little to inequalities because of compensating differentials, idiosyncratic preferences account for about 20% of overall inequality, which suggests workers benefit from information rents.

Labor market mobility in the competitive equilibrium is not constrained efficient and at least one third of job-to-job transitions are inefficient. To highlight the novelty of our results, we distinguish between general-equilibrium and partial-equilibrium inefficiencies, and focus on the latter: whether workers' job acceptance decisions are efficient, taking aggregate labor market tightness as given.[2] The inefficiency arises because when deciding whether to accept a job, a worker internalizes only her rent from the match, which varies across jobs, while the planner compares total match surplus. As a result, workers may choose jobs with lower total surplus if those jobs allow them to capture larger rents. Specifically, workers may leave jobs they like to avoid revealing their attachment to the incumbent employer, which would weaken their future bargaining position. Quantitatively, we find that the equilibrium features too much employer-to-employer mobility: at least one third of job-to-job transitions are inefficient. Although worker mobility across jobs is often interpreted as welfare-enhancing, our results show that a substantial share of these transitions can reduce surplus.

Finally, we study the effects of salary history bans, policies that restrict prospective employers from asking workers about their current or past wages. These policies, adopted in several U.S. jurisdictions and included in the 2023 EU Pay Transparency Directive, are intended to weaken the link between past and future wages and reduce inequality: if a worker is underpaid today, a future employer should be less able to use this information to perpetuate the initial disadvantage. The model provides a natural laboratory to study these policies because firms have incomplete information about worker preferences and value information contained in workers' labor-market histories. By restricting this information, salary history bans limit firms' ability to price discriminate across workers. Quantitatively, the model implies that these policies raise average wages by about 5% and reduce wage inequality: the gap between the 90th and 10th percentiles shrinks by 10 percentage points. At the same time, the policy reduces allocative efficiency: limiting firms' ability to price discriminate across workers worsens allocative efficiency and reduces average productivity.

[2]The general-equilibrium inefficiency is standard and concerns whether vacancy-posting decisions deliver the efficient level of labor market tightness. The partial-equilibrium inefficiency is more novel and concerns whether workers choose the efficient job conditional on the offers they receive.

Related literature. This paper relates to several strands of the literature. First, it contributes to the literature studying the role of worker preferences for jobs in frictional labor markets (Brügemann and Moscarini, 2010; Albrecht, Carrillo-Tudela, and Vroman, 2018; Taber and Vejlin, 2020; Lindenlaub and Postel-Vinay, 2023; Berger et al., 2023; Sullivan and To, 2014; Faberman, Mueller, and Şahin, 2026; Bagga et al., 2025; Lamadon et al., 2024; Morchio and Moser, 2026). One strand of this literature focuses on allocative efficiency and the horse race between assigning workers to more productive jobs and assigning them to jobs they value. A second strand uses frictional labor market models and matched employer–employee data to infer the size of worker preferences for jobs and the distribution of rents in the labor market.

Our paper extends this literature by decomposing non-wage job value into an observable firm-specific component and an unobservable worker-specific component. This distinction has two key implications. First, because worker-specific preferences are private information, they generate dynamic monopsony power and distinguish our model from Lamadon et al. (2024) and Berger et al. (2023). Firms learn about workers' preferences over time from their mobility decisions, and this learning generates heterogeneous rents across jobs. As a result, workers may switch jobs not because the new match generates higher surplus, but because it allows them to capture larger rents, leading to inefficient and excessive job-to-job mobility. Moreover, incomplete information makes workers' labor market histories valuable to employers, making the model a natural framework to study policies, such as salary history bans, that restrict the information available to prospective employers. Second, the model shows that observable and unobservable non-wage amenities have different implications for wage dispersion and inequality: observable amenities generate compensating differentials, while unobserved idiosyncratic preferences generate private rents for workers. Thus, our paper shows that if worker preferences for jobs are unobserved, as argued by Card (2022), they have very different implications for labor market dynamics than if they are observed.

Second, this paper relates to a large literature on monopsony power in labor markets (Manning, 2003; Card et al., 2018; Berger, Herkenhoff, and Mongey, 2022; Jarosch, Nimczik, and Sorkin, 2024; Jungerman, 2024; Volpe, 2024; Kline, 2025; Cheremukhin and Restrepo-Echavarria, 2025; Agostinelli et al., 2025). The idea that workers' private information can generate labor market power dates back to Robinson (1933), and has been studied extensively in models of monopsonistic and oligopsonistic labor markets. However, as emphasized by Berger et al. (2023), the literature lacks a framework to study labor market power in dynamic frictional labor markets. Existing models in which workers' preferences for jobs are private information are essentially static and therefore are silent on wage dynamics and job-to-job mobility. Our paper aims to fill this gap.

Recent dynamic models of monopsony include Jungerman (2024), Bloesch, Larsen, and Yding (2024), Agostinelli et al. (2025). Jungerman (2024) emphasizes the role of human capital accumulation. Bloesch, Larsen, and Yding (2024) focuses on how market power changes over the life-cycle of firms. Agostinelli et al. (2025) study how long-term employment relationships and dynamic wage contracts affect the quantification of labor market power. By contrast, our paper studies how preferences influence worker earnings and labor market dynamics. To this end, we extend the standard incomplete-information model of monopsony to a search framework where firms endogenously update their beliefs about those preferences. This comes with key implications for individual wage dynamics and worker mobility.

The dynamic nature of our model implies that firms learn about worker preferences over time because of their mobility decision. This form of learning is distinct from those already studied in the literature because it is asymmetric between firms and workers: firms always learn about preferences after workers. This asymmetry is critical for the determination of wages and, as a result, the mobility decisions of workers. By contrast, in Jovanovic (1979) both firms and workers learn together about the match productivity so the model has very different implications for wages and worker mobility. A recent quantitative literature argues that learning about productivity is critical for labor market dynamics (Papageorgiou, 2014; Gervais et al., 2016; Menzio, Telyukova, and Visschers, 2016; Bradley and Mann, 2024). We show that learning about preferences for jobs is also critical.

Finally, this paper complements empirical work on compensating differentials and worker willingness to pay for job attributes (Eriksson and Kristensen, 2014; Wiswall and Zafar, 2018; Hall and Mueller, 2018; Maestas et al., 2023; Lavetti, 2023; Mas, 2025) by studying the equilibrium implications of these preferences within a dynamic model of the labor market. We also provide new estimates of worker preferences from job-to-job transitions using administrative data and show that they are consistent with those from survey data.

The remainder of the paper is organized as follows. Section 3 presents the model. Section 4 analyzes the mechanism and characterizes the sources of inefficiency. Section 5 presents the quantitative results and discusses their aggregate implications.

2 Motivating Evidence on Job-to-Job Transitions

We use administrative data from France to document that worker preferences drive employer-to-employer (EE) mobility, and that these preferences are heterogeneous across workers. We do this by showing the prevalence of EE transitions in which workers receive wage cuts, and in which workers switch to firms that are less desirable to other workers. We also show that these EE transitions with lower wages and lower firm values cannot be fully accounted for by

alternative explanations such as layoffs, godfather shocks or higher expected wage growth at the next job. Instead, we provide suggestive evidence that these moves correlate with non-wage characteristics of jobs, such as proximity between home and work. Taken together, this new evidence from realized EE transitions is consistent with recent survey data (Mas and Pallais, 2017; Maestas et al., 2023; Caldwell, Haegele, and Heining, 2025), which shows that workers' willingness to pay for non-wage job characteristics is large and heterogeneous.

2.1 Matched Employer-Employee Data

We combine data on the universe of worker employment history from social security data (DADS) to annual information on firm balance sheet (FIBEN) in France between 1995 and 2019. We focus on full-time workers between the age of 25 and 55 working at private-sector for-profit firms. The data includes approximately 700,000 workers and 100,000 firms. Throughout, we define wages as total labor earnings divided by the number of hours and an EE transition when a worker switches employer within 2 weeks.

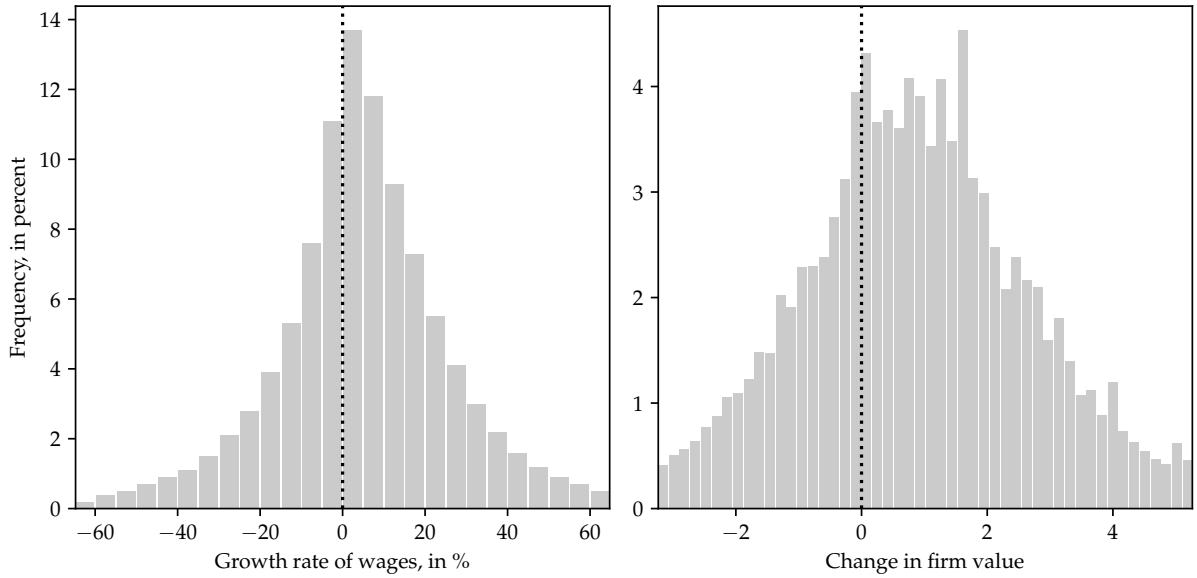
2.2 Worker preferences drive EE mobility

In France, 40% of workers receive wage cuts during EE transitions. This can be seen from the left panel of figure 1, which shows the distribution of wage changes during EE transitions. We complement this well known fact (Jolivet, Postel-Vinay, and Robin, 2006) with new evidence that these EE transitions with wage cuts cannot be fully accounted for by layoffs, godfather shocks or higher expected wage growth at the next job. We then show suggestive evidence from commuting distance between home and worker that non-wage characteristics of jobs improve during these EE transitions.

Transitions due to layoffs. One explanation for these transitions is that workers are actually laid off and move to the first job they find, potentially incurring a wage cut. We show that there is evidence of this in our data, but it only accounts for a minority of EE transitions with wage cuts. The results of this exercise are shown in table 2.

Our data allows us to assess whether layoffs are likely because the worker's existing employer is doing poorly. For example, we find that workers are 2 percentage points more likely to experience wage cuts during EE transitions if their previous employer experiences negative productivity or employment growth. Workers are also more likely to switch jobs if they expect their wages to fall in the future at the current firm, even if they are not laid off. Consistent with this explanation, we find that workers are 8 percentage points more likely to receive a wage cut during an EE transition if their co-workers from the previous

Figure 1: Changes in wages and firm values during EE transitions



Share of EE transitions	Full sample	$\Delta x \geq 0$, $\Delta n \geq 0$	$\Delta w \geq 0$ past co-workers	Permanent contracts	Same hours and location	Lower Δw post EE
With cuts in wages	38%	36%	37%	38%	35%	32%
With cuts in firm value	31%	31%	33%	30%	30%	29%

Table 1: Accounting for wage cuts during EE transitions without preferences

employer who did not switch jobs experience negative wage growth. However, Table 2 shows that these explanations do not account for all wage cuts during EE transitions. In fact, even if we focus on firms where co-workers experience positive wage growth, or where the firm experiences positive productivity or employment growth, almost as many workers receive wage cuts when they switch jobs.

Some layoffs are driven by worker-specific performances and are thus more challenging to rule out than layoffs affecting all workers within the firm. We leverage a key feature of the French labor market to further investigate whether such idiosyncratic layoffs can account for EE transitions with wage cuts. Specifically, we focus on a sample of workers with permanent contracts and more than 4 months of seniority. These workers are very difficult to lay off in France without cause and when the firm is doing well overall. Among these workers, we find that 40% still experience wage cuts when they switch jobs. This confirms that layoffs are not a main driver of such EE transitions, even when they are worker-specific.

Transitions due to godfather shocks. Workers might also experience life events forcing them to quit and take the first job they find. These events are comonly known as godfather shocks in the literature (Jolivet, Postel-Vinay, and Robin, 2006). While these can be interpreted as preference shocks, they do not necessarily fit the preferences we model in this paper (e.g. constant over time, specific to a job). Two common justifications for godfather shocks are relocation due to a spouse or the need to take some time off to take care of a newborn or family member. To assess whether these explanations account for the majority of EE transitions with wage cuts, we focus on a sub-sample where the new job is within the same commuting area as the previous job, and is a full-time job. We confirm that X% of workers in this sub-sample still experience wage cuts when they switch jobs, which suggests that these specific types of godfather shocks play a limited role.

Transitions to jobs with higher wage growth. Another explanation for wage cuts is that workers move to jobs where wages grow faster so that the present value of wages increases during EE transitions, even though current wages fall. This is a standard prediction of search models with firm heterogeneity where firms that are more productive offer more backloaded compensation (Burdett and Coles, 2010) Here again, we find evidence consistent with this explanation. In particular, workers receiving wage cuts during transitions are 8 percentage points more likely to experience higher wage growth at the next job than at their previous employer. However, column 5 in Table 2 shows that even if we focus on workers who do not experience higher wage growth at the next job, 35% of workers still receive wage cuts. Thus, the idea that workers accept lower wages today because they expect the present value of wages to increase cannot entirely account for EE transitions with wage cuts. Besides, in this another challenge to this explanation is that workers experiencing wage cuts are more likely to move to less productive firms in our data, not more likely as implied by this theory.

Evidence on worker preferences. We use information contained in our administration data on workers' location of worker and residence to compute their commuting distance and then evaluate how this distance changes during EE transitions. We find that workers reduce their commute by 0.4% when they switch jobs relative to the average commuting distance, which is fairly modest. However, this number varies significantly for workers experiencing wage cuts or wage increases during transitions. Workers in the lowest decile for the wage change distribution reduce their commute by 2.5% while workers in the top decile increase their commute by 3%. This evidence is consistent with existing work showing that worker satisfaction at their job predicts EE mobility.

Share of EE transitions	Full sample	$\Delta x \geq 0$, $\Delta n \geq 0$	$\Delta w \geq 0$ past co-workers	Permanent contracts	Same hours and location	Lower Δw post EE
With cuts in wages	38%	36%	37%	38%	35%	32%
With cuts in firm value	31%	31%	33%	30%	30%	29%

Table 2: Accounting for wage cuts during EE transitions without preferences

2.3 Worker preferences are heterogeneous

We now document that the preferences driving EE mobility are heterogeneous across workers. We reach this conclusion by first constructing a ranking of firms following the revealed preference methodology of Sorkin (2018) and then computing the change in firm value during each EE transitions. The distribution of such change in firm values is shown in the right panel of figure 1. We find that 31% of workers move to firms with lower value. This number would be closer to 0% if all workers had the same preferences for jobs, whereas it would be close to 50% if preferences were the only driver of job-to-job mobility. Our estimates thus indicate a significant degree of heterogeneity in preferences across workers. This result is critical for our analysis because it determines how preferences influence wages. Indeed, preferences can be inferred by firms when they are common across workers, and thus priced into wages, whereas they cannot be inferred directly when they are heterogeneous.

Before analyzing this implication further through the lens of our model, we address some concerns that might arise with our interpretation of the data. We show that layoffs, godfather shocks or differences in match productivity are unlikely to account for the bulk of EE transitions with cuts in firm values. We also show that heterogeneity in preferences based on observable worker and job characteristics (age, gender, income or year), while present in the data, does not influence worker mobility.

Layoffs, godfather shocks The second line of table 2 replicates the analysis of section 2.1 for changes in firm value during EE transitions. Across all columns, the share of EE transitions in which workers move to less desirable firms changes very little. This indicates that layoffs or godfather shocks arising from relocation or a need for flexible hours do not account for the bulk of these transitions.

Match productivity A competing explanation for these transitions down the job ladder is that workers move to jobs with higher match productivity. This increase in productivity

	Sex	Age: ±35	Top vs bottom income dist.	Pre vs post 2010
Correlation in rankings across groups	0.65	0.62	0.62	0.54
Share of EE transitions with value cuts	70%	67%	69%	70%

Table 3: Heterogeneity in job ladders based on observable characteristics.

would not be reflected in the firm value because it is specific to a match, and it would thus lead to transitions down the job ladder. It would, however, be reflected in the change in wages that workers experience during these transitions. One would expect workers to be more likely to receive a wage increase while moving to firms down the job ladder if these transitions are driven by match productivity. Otherwise, workers would not have any incentives to move to more productive matches. In our data, we find very little evidence of this. Workers are actually 4 percentage points more likely to move to a firm with higher value if they also receive a wage increase, and not the opposite. Accounting for wage growth at the next job gives more support to the idea that match productivity drives these EE transitions: workers are 4 percentage points more likely to move down the job ladder if they experience higher wage growth at the next job. However, this effect is small: even among workers experiencing lower wage growth after EE transitions, 29% move to firms with lower values.

Heterogeneity in job ladders based on observable characteristics. Another concern is that the heterogeneity estimated in the data is not worker-specific but rather connected to observable characteristics of workers or jobs. For example, we could observe transitions down the job ladder because rich and poor workers disagree on the ranking of firms. This is important because these characteristics are observable to firms, so they could in principle be priced into wages. To assess the importance of this type of observable heterogeneity and its implications for EE transitions, we estimate the ranking of firms on sub-samples of the data: men vs. women, young vs. old, rich vs. poor, and on a sample pre-2010 vs post-2010 to allow for changes in rankings over time. The first line of table 3 shows the correlation between the rankings across groups. There is significant heterogeneity based on observable characteristics in the rankings: the correlation is at best 0.65, and sometimes as low as 0.5. The second line of table 3 shows the share of workers moving to firms with lower value when the job ladder is estimated separately for each group. We find that the share of workers moving down the job ladder is almost identical than our baseline estimate of 68%, suggesting that the heterogeneity in preferences based on observable characteristics matters little for worker

mobility across jobs.

3 Model

We now develop a model to study how heterogeneous preferences for jobs shape wages, markdowns, and labor market flows. The model combines a dynamic search environment with on-the-job search and a source of monopsony power: workers have heterogeneous preferences for jobs that are not observed by firms. Firms therefore set wages based on beliefs about workers' preferences, and these beliefs evolve endogenously as workers accept, reject, or leave jobs. We begin by describing the environment, agents, and information structure. We then characterize workers' and firms' problems, describe wage setting for unemployed and employed workers, formalize belief updating, and conclude with the equilibrium definition.

3.1 Environment

Time is discrete and indexed by $t = 0, 1, 2, \ldots$. The economy is populated by a unit mass of risk-neutral workers and a large mass of risk-neutral firms. Workers discount future utility at rate $\beta \in (0, 1)$. Firms operate a constant-returns-to-scale technology and are heterogeneous in productivity $z \in \mathcal{Z}$ and in the value of a non-wage amenity $a \in \mathcal{A}$. Both z and a are fixed over time for a given firm and observed by all agents. Output is linear in labor: a firm with productivity z employing one worker produces z units of output per period.

The labor market is frictional, as in Postel-Vinay and Robin (2002). Workers can be unemployed or employed. Unemployed workers search for jobs and may receive job offers. Employed workers search on the job and may receive outside offers, allowing for direct employment-to-employment (EE) transitions. Matches are subject to an idiosyncratic separation shock, which occurs with probability $\delta \in (0, 1)$.

Workers and firms meet through random search. Firms post vacancies at flow cost $\kappa > 0$. Unemployed workers search with unit efficiency, while employed workers search with efficiency $s \in (0, 1)$. Let u denote the measure of unemployed workers. The effective measure of searching workers is therefore $u + s(1 - u)$. Given a measure v of vacancies, the number of new meetings is determined by a constant-returns-to-scale matching function

$$M\big(v, u + s(1 - u)\big).$$

Search is random: workers do not direct their search toward specific firms, and firms cannot target specific workers. Thus, conditional on a meeting, the worker–firm pair is drawn at random.

This search technology implies different offer-arrival rates for unemployed and employed workers, as well as a vacancy-filling rate for firms. Let $\lambda(\theta)$ denote the rate at which unemployed workers receive job offers, and let $\lambda_f(\theta)$ denote the rate at which firms fill vacancies. Since employed workers search with efficiency s, they receive outside offers at rate $s\lambda(\theta)$.

Worker preferences and information. A central feature of the model is that workers have idiosyncratic preferences for jobs. When worker i accepts a job at firm j, she draws a match-specific preference ξ_{ij}. This preference is persistent for the duration of the employment relationship.[3] We interpret ξ_{ij} broadly as the value worker i assigns to non-wage characteristics of the job that are specific to her match with firm j. These include, for example, interactions with coworkers and supervisors, the pace and intensity of work, scheduling flexibility, firm culture, organizational practices, and the overall work environment. Recent evidence documents substantial dispersion in how much workers value these attributes (Maestas et al., 2023).

Preferences are drawn from a common distribution $B_0(\xi)$.[4] Workers observe their own preference ξ_{ij}, but firms never observe it directly. Moreover, as we discuss below, workers have no incentive to reveal their preferences truthfully. Firms therefore make wage offers based on beliefs about ξ_{ij}, summarized by a distribution $B(\xi)$. These beliefs evolve over time as firms observe workers' mobility decisions.

We assume that workers draw their job-specific preference only after accepting a job offer. This assumption captures the idea that jobs are experience goods: workers learn how much they value a job only after trying it (Menzio and Shi, 2011). It also makes the learning problem tractable, since firms begin each new employment relationship with the common prior B_0 and update beliefs only through subsequent mobility decisions.

The flow utility of worker i employed at firm j is

$$w_{ij} + a_j + \xi_{ij},$$

where w_{ij} is the wage, a_j is the value of the firm-specific non-wage amenity, and ξ_{ij} is the worker-specific preference for the job. Thus, the value of non-wage amenities has two components: a common firm-level component a_j and an idiosyncratic match-specific component ξ_{ij}. The firm-level component is observed by all agents, while the worker-specific component is observed only by the worker.

[3] This assumption differs from much of the literature, which assumes that preferences are i.i.d. over time, mostly for tractability.

[4] In the quantitative analysis, we assume a parametric form for B_0, but this is not required for the theoretical characterization.

Wages. We extend the bargaining protocol of Postel-Vinay and Robin (2002) to a setting with private information about workers' preferences. Because search is random, a firm that posts a vacancy can meet either an unemployed worker or an employed worker. The wage-setting protocol differs across these two cases.

First, consider a firm that meets an unemployed worker. The firm makes a take-it-or-leave-it wage offer. If preferences were observed by the firm, the firm could tailor the wage to the worker's exact outside option and match-specific preference. In our environment, however, the worker's preference for the job is private information. The firm therefore does not know how much the worker values the match and chooses a wage given its belief about the worker's preference. As a result, the firm faces an upward-sloping labor supply curve: a higher wage raises the probability that the worker accepts the offer, but this probability generally lies strictly between zero and one. Thus, even though the firm makes a take-it-or-leave-it offer, it cannot extract the entire match surplus from every worker. Some workers receive rents because the firm cannot offer a wage that makes all possible preference types exactly indifferent between accepting the job and remaining unemployed.

Second, consider a firm that meets an employed worker. In this case, the worker's current employer and the poaching firm compete for the worker. We assume that the incumbent and the poacher make wage offers à la Bertrand given their beliefs about the worker's preference for the current job. The worker then chooses whether to remain with the incumbent or move to the poacher. Private information again matters because the worker's preference for the incumbent employer is observed by the worker but not by either firm. Hence, firms do not compete for a worker with known preferences; they compete given beliefs about how attached the worker is to the current job. These beliefs affect both the probability that a given wage offer attracts or retains the worker and the value of the match going forward.

Firms make offers in terms of the per-period wage, which they are assumed to honor. Wages adjust only in response to outside offers, and the wage of an incumbent worker cannot fall following an outside offer. This assumption is standard in the literature (Cahuc, Postel-Vinay, and Robin, 2006; Lise and Robin, 2017; Jarosch, 2023) and generates wage dynamics consistent with the data. We present the firms' wage-setting problems formally in Section 3.

Timing. Within each period, unemployed and employed workers first search for jobs, while firms post vacancies. Meetings between workers and firms are then realized according to the random search process. After a meeting occurs, firms make wage offers, and workers decide whether to accept the offer or remain in their current state. If an employed worker receives an outside offer, the incumbent and poaching firms simultaneously make wage offers. The worker chooses the firm that maximizes her expected utility. For unemployed workers, this

decision is whether to accept the job offer or remain unemployed. For employed workers, this decision is whether to stay with the incumbent employer or move to the poaching firm. Production then takes place, wages are paid, and existing matches are subject to exogenous separation shocks.

3.2 Value Functions and Beliefs

We illustrate the problem of workers and firms by characterizing their value functions, and we describe the law of motion of beliefs. The main decision problem each worker faces is whether to accept or not a job offer, in case she receives one. Firms post vacancies and choose wages to maximize profits.

Unemployed workers. Let U denote the value of being unemployed. An unemployed worker receives a flow payoff $\bar{y}$, interpreted as home production or unemployment benefits. The worker receives job offers at rate $\lambda(\theta)$, where θ denotes aggregate labor market tightness.

The value of unemployment satisfies

$$U = \bar{y} + \beta\left[1 - \lambda(\theta)\right] U + \beta\lambda(\theta)\mathbb{E}_{z,a,\xi}\left[\max\left\{V\left(w_U(z,a), z, a, \xi, \mathcal{B}'_U\left(w_U(z,a)\right)\right), U\right\}\right]. \quad (1)$$

The unemployed worker enjoys flow utility $\bar{y}$ today. With probability $1 - \lambda(\theta)$, she does not receive a job offer and remains unemployed next period, yielding continuation value U. With probability $\lambda(\theta)$, she meets a firm with productivity z and amenity a. The firm makes a take-it-or-leave-it wage offer $w_U(z,a)$, and the worker draws a match-specific preference ξ for that job. She then chooses whether to keep the job and obtain value

$$V\left(w_U(z,a), z, a, \xi, \mathcal{B}'_U\left(w_U(z,a)\right)\right)$$

or to reject the offer and remain unemployed.

The continuation value from accepting the job depends on the belief $\mathcal{B}'_U$ that firms hold after observing that the worker accepted the offer. This posterior belief depends on the offered wage because accepting a low wage is more informative about the worker's preference for the job than accepting a high wage. We describe the law of motion of beliefs at the end of this section. Since search is random, the ex-ante value of meeting a firm is an expectation over firm types and match-specific preferences. Ex post, however, the value of accepting the job depends on the realized productivity z, amenity a, wage offer $w_U(z,a)$, preference draw ξ, and posterior belief $\mathcal{B}'_U\left(w_U(z,a)\right)$.[5]

[5]In our model, jobs are experience goods: workers observe the match-specific preference ξ immediately

Employed workers. Consider a worker employed at a firm with productivity z, amenity a, and wage w. Let $V(w, z, a, \xi, \mathcal{B})$ denote the value of employment when the worker's preference for the current job is ξ and firms hold belief $\mathcal{B}$ about this preference.

The value of employment satisfies

$$\begin{aligned} V(w, z, a, \xi, \mathcal{B}) = w + a + \xi + \beta\delta U \\ + \beta(1-\delta)\left[1 - s\lambda(\theta)\right] V(w, z, a, \xi, \mathcal{B}) \\ + \beta(1-\delta)s\lambda(\theta)\mathbb{E}_{z',a'}\left[\max\left\{V_I(w_I, w_P, z, a, \xi),\ V_P(w_P, z', a')\right\}\right], \end{aligned} \tag{2}$$

with continuation values V_I, V_P defined as:

$$V_I(w_I, w_P, z, a, \xi) \equiv V\left(w_I, z, a, \xi, \mathcal{B}'(w_I, w_P)\right), \tag{3}$$

$$V_P(w_P, z', a') \equiv \mathbb{E}_{\xi'}\left[\max\left\{V\left(w_P, z', a', \xi', \mathcal{B}'_U(w_P)\right), U\right\}\right]. \tag{4}$$

The first line gives the worker's current flow utility, which is equal to the wage w, the firm-specific amenity a, and the worker's idiosyncratic preference ξ for the current job. With probability δ, the match is exogenously destroyed and the worker becomes unemployed next period, obtaining value U. Conditional on survival, the worker does not receive an outside offer with probability $1 - s\lambda(\theta)$ and remains in the current job with continuation value $V(w, z, a, \xi, \mathcal{B})$. With probability $s\lambda(\theta)$, the worker receives an outside offer from a firm with productivity z' and amenity a'.

When an outside offer arrives, the incumbent firm and the poaching firm make wage offers w_I and w_P. The worker then chooses whether to stay with the incumbent or join the poacher. If she stays, her preference for the incumbent remains ξ, but firms update their belief about this preference to $\mathcal{B}'(w_I, w_P)$. This posterior belief depends on both wage offers because the decision to stay is informative about the worker's attachment to the incumbent only relative to the wage offered by the incumbent and the poaching firm.

If the worker joins the poacher, she draws a new match-specific preference ξ' for the poaching firm. The relevant continuation belief is then $\mathcal{B}'_U(w_P)$, which summarizes the belief about the worker's preference for the new employer after observing that she accepted the poacher's wage offer. This belief depends on the poacher's wage offer w_P. Even though the preference ξ' is drawn only after the worker accepts the job, the worker can immediately quit into unemployment if the value of the new match is below U. Therefore, acceptance and non-quit behavior are informative about the worker's preference for the poaching firm.

after joining the firm. Since workers can quit, the timing of this information is immaterial. Observing ξ before match formation would lead workers with sufficiently low draws to reject the offer; observing ξ immediately after match formation leads those same workers to accept the offer and immediately separate.

We describe the belief-updating rules formally at the end of this Section.

Value of a filled job. We now turn to the decision problem of firms. We describe firms' problem by characterizing the value of a filled job. Since production is linear in labor and firms can offer different wages to different workers, a firm can be viewed as a collection of jobs.

Let $\Pi(w, z, a, \mathcal{B})$ denote the value of a filled job for a firm with productivity z and amenity a, paying wage w, and holding belief $\mathcal{B}$ about the worker's preference for the current job. The value of a filled job satisfies

$$\Pi(w, z, a, \mathcal{B}) = z - w + \beta(1-\delta)\left[(1 - s\lambda(\theta))\,\Pi(w, z, a, \mathcal{B}) + s\lambda(\theta)\Pi^I(w, z, a, \mathcal{B})\right]. \tag{5}$$

The firm earns flow profit $z - w$ from employing the worker. With probability δ, the match is exogenously destroyed and the continuation value is zero.[6] With probability $1-\delta$, the match survives. Conditional on survival, the worker does not receive an outside offer with probability $1 - s\lambda(\theta)$, in which case the firm continues with value $\Pi(w, z, a, \mathcal{B})$. With probability $s\lambda(\theta)$, the worker receives an outside offer. The incumbent and the poaching firm then compete for the worker by making wage offers. The competition is Bayesian Nash, since optimal wage offers solve a game with incomplete information.

Wage setting for employed workers. Firms simultaneously submit wage offers to the worker. Although the worker's preference for the current job is private information, all actions are observed. Hence, both firms share the same belief $\mathcal{B}$ about the worker's preference for the incumbent employer when they choose their wage offers. The worker's decision then determines whether she remains with the incumbent or joins the poacher, and this decision updates beliefs.

Let

$$P(\text{stay} \mid w_I, w_P, z_I, a_I, z_P, a_P, \mathcal{B})$$

denote the probability that the worker stays with the incumbent, given wage offers w_I and w_P, firm types (z_I, a_I) and (z_P, a_P), and belief $\mathcal{B}$ about the worker's preference for the incumbent. The incumbent chooses w_I to maximize the expected value of retaining the worker:

$$\Pi^I(w, z, a, \mathcal{B}) = \mathbb{E}_{z',a'}\left[\max_{w_I \geq w} P(\text{stay} \mid w_I, w_P, z, a, z', a', \mathcal{B}) \cdot \Pi\left(w_I, z, a, \mathcal{B}'(w_I, w_P)\right)\right]. \tag{6}$$

[6] This follows from the free-entry condition (9).

The incumbent's payoff depends on two terms. The first is the probability of retaining the worker, which is increasing in the wage offered by the incumbent and depends on the wage offered by the poacher. The second is the value of the job conditional on retaining the worker. If the worker stays, firms update their belief about the worker's preference for the incumbent to $\mathcal{B}'(w_I, w_P)$. This posterior belief depends on both wage offers because the decision to stay is informative only relative to the outside offer faced by the worker.

Similarly, the poaching firm chooses w_P to maximize the expected value of attracting the worker:

$$\Pi^P(z_I, a_I, z_P, a_P, \mathcal{B}) = \max_{w_P} \left[P(\text{accept} \mid w_I, w_P, z_I, a_I, z_P, a_P, \mathcal{B})\right] \cdot \Pi\left(w_P, z_P, a_P, \mathcal{B}'_U(w_P)\right). \tag{7}$$

The poacher's payoff also depends on two terms: the probability of attracting the worker and the value of the job conditional on hiring her. If the worker joins the poacher and does not immediately quit, the relevant posterior belief is $\mathcal{B}'_U(w_P)$. This belief concerns the worker's preference for the poaching firm, not her preference for the incumbent. It depends on the poacher's wage offer because, after accepting the offer, the worker draws a new match-specific preference for the poaching firm and can immediately quit into unemployment if the value of the new match is below U. Thus, accepting the poacher's offer and remaining in the job is informative about the worker's preference for the poaching firm.

The incumbent and poacher problems show that worker preferences can generate monopsony power even in a setting with direct competition between two firms. This is because preferences are private information. Firms do not know how much the worker values the incumbent job relative to the poaching job, so each wage offer changes the probability of attracting or retaining the worker rather than determining the outcome with certainty. In this sense, firms face upward-sloping labor supply curves, captured by the probabilities of successfully attracting or retaining the worker.

Evolution of beliefs. Beliefs evolve according to Bayes' rule. Observing a worker's decision to stay or leave conveys information about the worker's preference ξ for the incumbent employer. Formally, a belief $\mathcal{B}$ is a distribution over values of ξ, and this distribution is updated according to

$$\mathcal{B}'(\xi) = \frac{\mathbb{1}\{\text{stay} \mid \xi, w_I, w_P, z_I, a_I, z_P, a_P\}\mathcal{B}(\xi)}{\int \mathbb{1}\{\text{stay} \mid \tilde{\xi}, w_I, w_P, z_I, a_I, z_P, a_P\}\mathcal{B}(\tilde{\xi})d\tilde{\xi}}. \tag{8}$$

The informativeness of the worker's decision depends on wage offers, productivities, and amenities. When w_I is much larger than w_P, staying reveals little about preferences; when

the offers are close, staying is highly informative. Similarly, a worker may keep her current job when $w_I < w_P$, but this decision may reveal little about ξ if the incumbent job has higher productivity or better amenities. In that case, the worker may stay because the incumbent job is otherwise more attractive, not because she has a strong idiosyncratic preference for it. Finally, observing a worker keep her job when $w_I < w_P$ may reveal little additional information about ξ if the prior belief $\mathcal{B}$ is already heavily skewed toward high values of ξ.

3.3 Equilibrium

Free entry in the labor market implies that firms post vacancies until the expected value of entry equals the vacancy posting cost. Since search is random, a vacancy can be filled either by an unemployed worker or by an employed worker. Thus, the free-entry condition is

$$\lambda_f(\theta)\left[\frac{u}{u+s(1-u)}\mathbb{E}_{z,a}\left[\Pi_0^U(z,a)\right]+\frac{s(1-u)}{u+s(1-u)}\mathbb{E}_{z_I,a_I,z_P,a_P,\mathcal{B}}\left[\Pi^P(z_I,a_I,z_P,a_P,\mathcal{B})\right]\right]=\kappa, \tag{9}$$

where u denotes the unemployment rate. The first term inside the brackets is the expected value of filling a vacancy with an unemployed worker. The second term is the expected value of filling a vacancy by poaching an employed worker from another firm. Since matching is random, firms cannot direct vacancies toward particular workers or firm types, and there is a single aggregate free-entry condition. We define an equilibrium as follows.

Definition 1. *An equilibrium consists of value functions $(U, V, \Pi, \Pi^I, \Pi^P)$, wage schedules (w_U, w_I, w_P), beliefs $\mathcal{B}(\xi)$, labor market tightness θ, and transition rates $(\lambda(\theta), \lambda_f(\theta))$ such that:*

1. *Workers' value functions satisfy equations (1) and (2).*
2. *Firms' value functions satisfy equations (5)–(7).*
3. *Wage offers maximize firms' profits given beliefs.*
4. *Workers' mobility decisions are optimal.*
5. *Beliefs evolve according to Bayes' rule (8).*
6. *The free-entry condition (9) holds.*

3.4 Discussion

We now discuss three modeling choices about the structure of non-wage preferences and the information available to firms.

Our model assumes that workers derive utility from non-wage characteristics of jobs. We represent this utility as the sum of two components: a firm-specific component a_j and a match-specific component ξ_{ij}. The firm-specific component captures amenities that are common across workers and observable to firms, such as formal work-from-home policies, benefits, scheduling rules, or other workplace policies. The match-specific component captures the part of job value that is specific to the worker–firm match and privately observed by the worker, such as how much the worker likes her team, her supervisor, the pace of work, or the day-to-day work environment. Thus, the flow utility of worker i employed at firm j is additive in wages, firm amenities, and match-specific preferences:

$$w_{ij} + a_j + \xi_{ij}.$$

This additive specification is not essential. An alternative formulation would allow the value of firm amenities to depend on worker-specific tastes. For example, worker utility could include a term such as $\xi_i a_j$, where a_j captures a firm-level policy, such as the possibility of working from home, and ξ_i captures how much worker i values that policy. Under this interpretation, firms differ in the amenities they provide, while workers differ in their willingness to trade wages for those amenities. Our framework could be extended to accommodate this multiplicative structure. The key mechanism would remain the same: firms would not perfectly observe workers' valuations of non-wage job characteristics and would therefore set wages based on beliefs.

Our baseline specification assumes that the match-specific preference ξ_{ij} is persistent throughout the employment relationship. This assumption implies that firms learn about a worker's attachment to the current job over the worker's tenure at that firm. An alternative specification would make the relevant preference parameter worker-specific rather than match-specific. For example, workers could differ permanently in how much they value wages relative to non-wage amenities, as in Volpe (2024). In that case, ξ_i would be a worker type that scales the value of amenities, or directly affects the marginal utility of wages. Firms would then learn not only about the worker's attachment to a particular job, but about the worker's broader preferences across jobs. The learning horizon would therefore be the worker's labor market experience rather than her tenure with a given employer. Our framework can be modified to accommodate this alternative environment.

Finally, in the baseline model, firms learn about preferences only from workers' mobility decisions. In practice, firms may also form beliefs using observable worker characteristics, such as age, gender, occupation, or labor market history. This can be incorporated by allowing different groups of workers to draw ξ from different distributions. Firms would

then enter a match with group-specific priors and update those beliefs using the worker's subsequent mobility decisions. This extension would preserve the main mechanism of the model while allowing beliefs to depend both on observable characteristics and on revealed preference through job-to-job mobility.

4 Model Mechanism

We characterize how worker preferences influence worker mobility and wages and illustrate the main model mechanisms. Section 4.1 shows that workers' mobility decisions follow a cutoff rule in the preference ξ, which simplifies the law of motion of beliefs under a parametric assumption on the distribution of preferences. Section 4.2 characterizes firms' optimal wage offers and discusses labor market dynamics, emphasizing how beliefs evolve over tenure and feed back into wages. Section 4.3 discusses efficiency and shows that the labor market flows generated by workers' mobility decisions are not constrained efficient.

4.1 Worker Mobility Decisions

Consider a worker currently employed at an incumbent firm with productivity z_I and amenity a_I, who receives an outside offer from a poaching firm with productivity z_P and amenity a_P. The incumbent offers wage w_I, while the poacher offers wage w_P. Let $V(w, z, a, \xi, \mathcal{B})$ denote the worker's value of employment at a firm with productivity z and amenity a, when the wage is w, the worker's match-specific preference is ξ, and firms hold belief $\mathcal{B}$ about this preference.

The worker stays with the incumbent if the value of remaining exceeds the value of switching. Since the preference for the incumbent job is private information, the stay/switch decision is monotone in ξ under a mild condition. First, define the reservation threshold ξ^* as the solution to

$$V\left(w_I, z_I, a_I, \xi^*, \mathcal{B}'(w_I, w_P)\right) = \mathbb{E}_{\xi_P}\left[\max\left\{V\left(w_P, z_P, a_P, \xi_P, \mathcal{B}'_U(w_P)\right), U\right\}\right]. \tag{10}$$

The left-hand side is the worker's value of staying with the incumbent when her preference for the incumbent job equals the threshold ξ^*. The reservation threshold is the value of ξ that makes the worker indifferent between staying with the incumbent and joining the poaching firm.

Importantly, the Bayesian Nash equilibrium that solves the decision problems of firms and workers requires that the updated belief $\mathcal{B}'$ in (10) be consistent with the cutoff rule ξ^*. This creates a fixed-point problem: firms make wage offers given the wage offered by the

other firm and the law of motion of beliefs; workers make mobility decisions given the wage offers of the two firms; and the law of motion of beliefs must be consistent with workers' mobility decisions.

The wage-setting problem of firms can be difficult to solve because beliefs about workers' preferences are infinite-dimensional state variables that themselves solve a fixed-point problem. The problem simplifies substantially when we assume that the distribution of preferences is uniform.[7]

Assumption 1. *The unconditional distribution of preferences $\mathcal{B}_0(\xi)$ is uniform with support*

$$[-\sqrt{3}\sigma_\xi, \sqrt{3}\sigma_\xi].$$

This assumption delivers a useful simplification. Under the cutoff rule above, observing whether the worker stays or switches truncates the support of the posterior distribution while preserving the Uniform functional form. As a result, beliefs can be summarized by a single scalar, the lower support of the posterior distribution, rather than by an entire distribution. Intuitively, when a worker stays after receiving a more attractive outside offer, the incumbent learns that her preference for the current job must be sufficiently high, shifting the posterior support upward.

Under this representation, the posterior belief can be summarized by its lower support. If the mapping

$$x \mapsto V\left(w_I, z_I, a_I, x, x\right)$$

is strictly increasing in x, then equation (10) pins down a unique cutoff ξ^* such that the worker's decision then follows a cutoff rule:

$$\text{stay with the incumbent} \quad \Longleftrightarrow \quad \xi \geq \xi^*(w_I, w_P, z_I, a_I, z_P, a_P, \mathcal{B}).$$

This monotonicity condition is natural: a higher match-specific preference for the incumbent makes staying more attractive. We restrict our attention to equilibria that satisfy this property and we verify it holds numerically.

A similar result applies to the decision of accepting a job from unemployment. Given an offer with wage w_U, productivity z, and amenity a, the worker follows a cutoff rule and joins the firm if $\xi \geq \xi^*_U$, where ξ^*_U satisfies

$$V(w_U, z, a, \xi^*_U, \xi^*_U) = U.$$

[7] The problem would also simplify under other functional-form assumptions, for example if the distribution of preferences followed a Pareto distribution.

We now establish a simple monotone comparative statics property of the reservation threshold. Holding firm types and beliefs fixed, workers are more likely to stay with the incumbent when the incumbent offers a higher wage or when the poacher offers a lower wage.

Lemma 1. *The reservation threshold $\xi^*(w_I, w_P, z_I, a_I, z_P, a_P, \mathcal{B})$ is decreasing in the wage offered by the incumbent w_I and increasing in the wage offered by the poacher w_P.*

Proof: see Appendix A.

4.2 Equilibrium Wage Dynamics

We now characterize optimal wage offers for incumbent and poaching firms. Wages depend on the productivity and amenity of the incumbent firm, (z_I, a_I), the productivity and amenity of the poaching firm, (z_P, a_P), and firms' belief about the worker's preference for the incumbent job. Under Assumption 1, this belief can be summarized by the lower support of the posterior distribution, denoted by $\underline{\mathcal{B}}$.

Let ξ^* denote the worker's reservation threshold, such that worker stays with the incumbent if $\xi \geq \xi^*$. Hence, the probability that the incumbent retains the worker is

$$P_I(w_I, w_P, z_I, a_I, z_P, a_P, \underline{\mathcal{B}}) = 1 - F\left(\xi^* \mid \underline{\mathcal{B}}\right),$$

where $F(\cdot|\underline{\mathcal{B}})$ is the cdf of a uniform distribution with lower support $\underline{\mathcal{B}}$ and same upper support as the unconditional distribution of preferences. The probability that the poacher attracts the worker depends on ξ^* and on the probability that the worker does not quit, which occurs if $\xi_P \geq \xi_P^*$, where ξ_P^* solves (11), where $F_0(\cdot)$ is the cdf of the unconditional distribution of preferences.

$$V(w_P, z_P, a_P, \xi_U^*, \xi_U^*) = U \tag{11}$$

$$P_P(w_I, w_P, z_I, a_I, z_P, a_P, \underline{\mathcal{B}}) = F\left(\xi^* \mid \underline{\mathcal{B}}\right)\left[1 - F_0(\xi_U^*)\right] \tag{12}$$

Incumbent firm. Consider first the incumbent firm. If the worker stays, the posterior belief about her preference for the incumbent job is updated to the new lower support ξ^*. Thus, the incumbent chooses w_I to maximize expected profits.

The optimal wage offered by the incumbent firm is implicitly characterized from the

optimality condition of the incumbent:

$$0 = \underbrace{\frac{\partial\left[1-F\left(\xi^{*} \mid \underline{\mathcal{B}}\right)\right]}{\partial \xi^{*}} \frac{\partial \xi^{*}}{\partial w_{I}} \Pi\left(w_{I}, z_{I}, a_{I}, \xi^{*}\right)}_{\text{retention}} + \underbrace{\left[1-F\left(\xi^{*} \mid \underline{\mathcal{B}}\right)\right] \frac{\partial \Pi\left(w_{I}, z_{I}, a_{I}, \xi^{*}\right)}{\partial w_{I}}}_{\text{profits}} + \underbrace{\left[1-F\left(\xi^{*} \mid \underline{\mathcal{B}}\right)\right] \frac{\partial \Pi\left(w_{I}, z_{I}, a_{I}, \xi^{*}\right)}{\partial \xi^{*}} \frac{\partial \xi^{*}}{\partial w_{I}}}_{\text{information}}. \tag{13}$$

Each term has a natural interpretation. The *retention* term captures that changing w_I changes the cutoff ξ^*, and therefore the probability that the worker stays with the incumbent. Since $\partial\xi^*/\partial w_I < 0$, a higher incumbent wage lowers the cutoff and raises the probability of retention. The *profits* term is the standard marginal cost of a higher wage, holding fixed the probability of retention and the composition of retained workers. The *information* term captures that changing w_I also changes which preference types stay. By lowering the cutoff, a higher wage allows the incumbent to retain workers with weaker attachment to the job, which affects future beliefs and therefore the continuation value of the match.

We now characterize how beliefs affect the incumbent's best response. All else equal, a higher value of $\underline{\mathcal{B}}$ means that the worker is believed to be more attached to the incumbent job. This lowers the retention elasticity faced by the incumbent and reduces the incentive to offer a high wage.

Proposition 1. *The incumbent best response $w_I^*(w_P; z_I, a_I, z_P, a_P, \underline{\mathcal{B}})$ is weakly decreasing in $\underline{\mathcal{B}}$, holding $(w_P, z_I, a_I, z_P, a_P)$ fixed.*

Proof: see Appendix A.

Poaching firm. The poaching firm earns positive value only if the worker switches. This occurs with probability $F\left(\xi^{*} \mid \underline{\mathcal{B}}\right)\left[1-F(\xi_U^*)\right]$. If the worker joins the poacher, she draws a new match-specific preference for the poaching firm. Since the worker can immediately quit into unemployment if the new match is sufficiently unattractive, the poacher's continuation value depends on the posterior belief $\mathcal{B}'_U(w_P)$, summarized by the lower support ξ_U^*.

The optimal wage offered by the poaching firm is implicitly characterized from the opti-

mality condition of the poacher:

$$0 = \underbrace{\frac{\partial \left(F\left(\xi^* \mid \underline{\mathcal{B}}\right)\left[1 - F(\xi_U^*)\right]\right)}{\partial \xi^*} \frac{\partial \xi^*}{\partial w_P} \Pi\left(w_P, z_P, a_P, \xi_U^*\right)}_{\text{poaching}} + \underbrace{F\left(\xi^* \mid \underline{\mathcal{B}}\right)\left[1 - F(\xi_U^*)\right] \frac{\partial \Pi\left(w_P, z_P, a_P, \xi_U^*\right)}{\partial w_P}}_{\text{profits}}$$
$$+ \underbrace{F\left(\xi^* \mid \underline{\mathcal{B}}\right)\left[1 - F(\xi_U^*)\right] \frac{\partial \Pi\left(w_P, z_P, a_P, \xi_U^*\right)}{\partial \xi_U^*} \frac{\partial \xi_U^*}{\partial w_P}}_{\text{information}} \tag{14}$$

The *poaching* term captures that raising w_P increases the probability of attracting the worker. By Lemma 1, a higher poaching wage raises the cutoff ξ^*, making the worker more likely to switch. The second term captures the direct effect of a higher wage on profits and the third term captures the effect of the wage on the posterior belief ξ_U^* about the worker's preference for the poaching firm.

We find that, all else equal, the poaching wage is increasing in the belief $\underline{\mathcal{B}}$. If the worker is believed to be strongly attached to the incumbent job, the poacher must offer a higher wage to attract her.

Proposition 2. *The poaching best response* $w_P^*(w_I; z_I, a_I, z_P, a_P, \underline{\mathcal{B}})$ *is weakly increasing in* $\underline{\mathcal{B}}$*, holding* $(w_I, z_I, a_I, z_P, a_P)$ *fixed.*

Proof: see Appendix A.

Best responses. Equilibrium wages are determined by the intersection of the incumbent and poacher best responses:

$$w_I = w_I^*(w_P; z_I, a_I, z_P, a_P, \underline{\mathcal{B}}), \qquad w_P = w_P^*(w_I; z_I, a_I, z_P, a_P, \underline{\mathcal{B}}).$$

This system determines how equilibrium wages depend on firm productivities, firm amenities, and beliefs about the worker's attachment to the incumbent job.

The belief $\underline{\mathcal{B}}$ affects equilibrium wages through two channels. First, changes in the beliefs affect the best responses of incumbent and poaching firms. For example, a higher belief means that the worker is more likely to value the incumbent job, which lowers the wage offered by the incumbent and increases the wage offered by the poacher. Second, these forces interact through *strategic interaction* in wage setting. The incumbent's optimal wage depends on the poacher's wage, and the poacher's optimal wage depends on the incumbent's wage. Therefore, shifts in one firm's best response feed back into the other firm's wage choice. The equilibrium effect of beliefs on wages combines direct monopsony power, competitive pressure from the poaching firm, and the strategic interaction between the two wage setters.

Figure 2 illustrates this logic. The figure shows how an increase in beliefs $\underline{\mathcal{B}}$ shifts both best-response functions and how equilibrium wages adjust as the intersection of these shifted schedules changes.

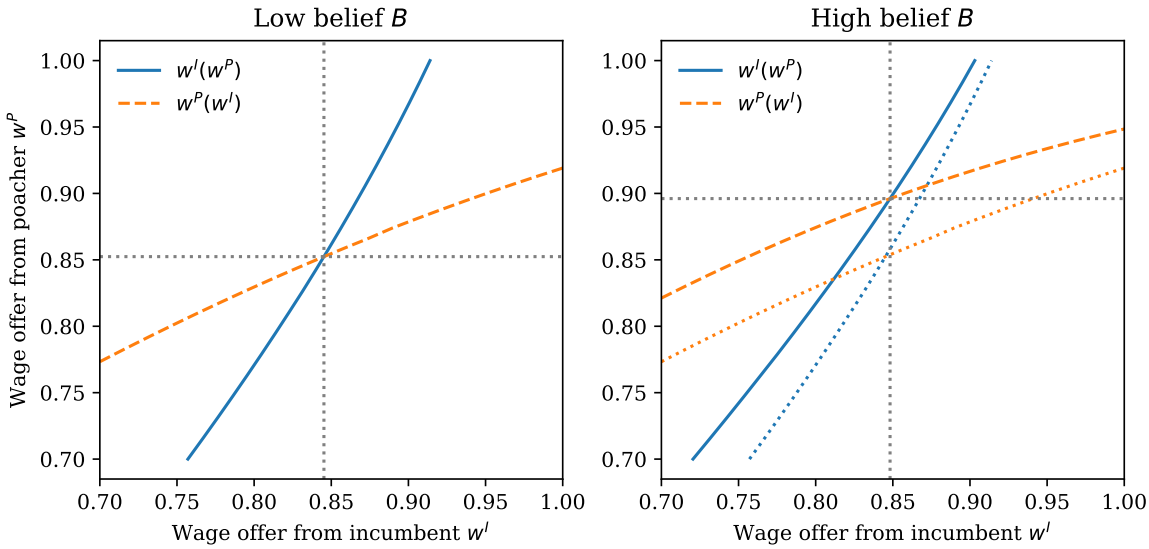


Figure 2: Equilibrium wages as the intersection of best responses.

Under the assumption that the distribution of preferences is Uniform, we can characterize how beliefs affect the equilibrium wage offered by the poacher.

Proposition 3. *The equilibrium wage offered by the poacher, w_P^*, is increasing in beliefs $\underline{\mathcal{B}}$.*

Proof: see Appendix A.

Labor market dynamics. We now illustrate how beliefs and wage offers evolve over the employment relationship. When a worker receives an outside offer, the incumbent may initially respond by raising the wage in order to retain the worker. These counteroffers generate wage growth on the job, as in Postel-Vinay and Robin (2002); Cahuc, Postel-Vinay, and Robin (2006). In our model, however, each rejected outside offer is also informative. If the worker stays, the incumbent learns that her preference for the current job exceeds the cutoff ξ^*, which shifts the posterior belief $\underline{\mathcal{B}}$ upward.

This learning changes the incumbent's response to future outside offers. As the worker repeatedly stays, the incumbent becomes increasingly confident that she is attached to the current job. As a result, the incumbent becomes less responsive to future outside offers: it knows that the worker is likely to stay even without a large wage increase. Thus, although outside offers initially induce wage increases, learning about the worker's attachment allows the incumbent to preserve labor market power over time.

This mechanism has important implications for markdowns. In Postel-Vinay and Robin (2002), repeated outside offers gradually bid up wages and markdowns converge toward one. In our model, this convergence is incomplete. Because staying reveals attachment to the incumbent job, firms learn that some workers can be retained without fully matching outside opportunities. Wage markdowns therefore remain high over the employment relationship instead of converging to one.

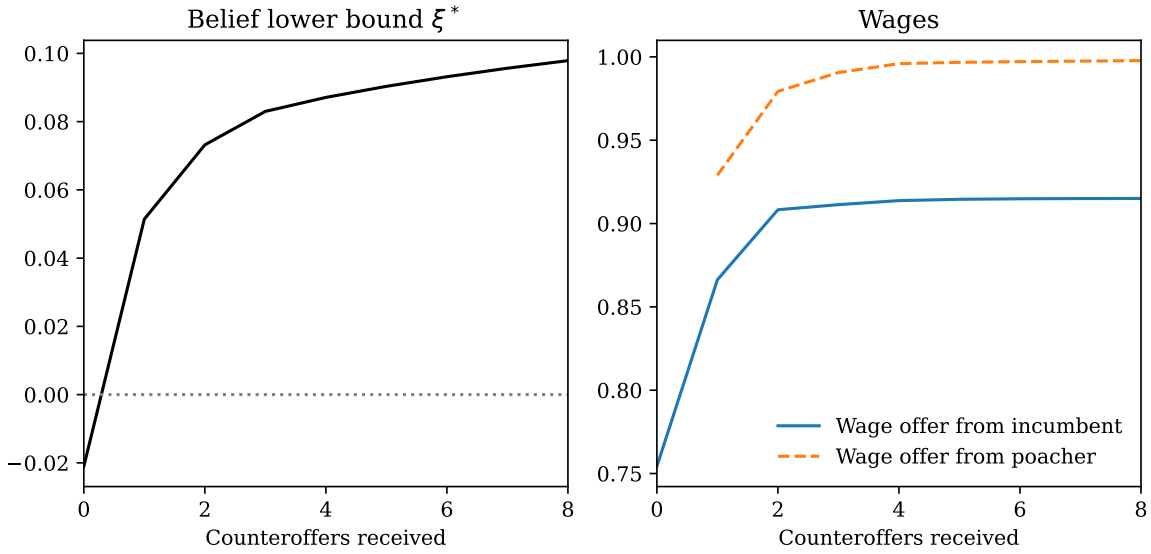


Figure 3: Belief and wage dynamics following counteroffers.

4.3 Inefficient Worker Mobility

We now characterize the constrained efficient allocation. We proceed in two steps. First, we characterize the problem and the optimal allocation of a social planner that observes worker preferences. We refer to this problem as the relaxed planner problem. Then, we show that such allocation can be implemented by a constrained planner that does not observe worker preferences.

Relaxed planner problem. The planner maximizes total surplus net of vacancy posting costs, taking as given the same search technology and information structure as in the decentralized economy, and the planner has the same information as workers regarding their preferences. The planner chooses labor market tightness and workers' mobility decisions. The planner problem is an extension of Fukui and Mukoyama (2025) to an economy with idiosyncratic worker preferences. This approach allows us to separate two sources of inefficiency. The first is a general-equilibrium inefficiency in vacancy creation, which determines

aggregate labor market tightness. The second is a partial-equilibrium inefficiency in workers' acceptance and mobility decisions, taking labor market tightness as given.

Let $D_t(z, a, \xi)$ denote the measure of workers employed at firms with productivity z, amenity a, and match-specific preference ξ at the beginning of period t. Let u_t denote unemployment, so that

$$u_t = 1 - \sum_{z,a,\xi} D_t(z, a, \xi).$$

The flow surplus generated by an employed match is $(z + a + \xi)$, while the flow value of unemployment is $\bar{y}$. If aggregate labor market tightness is θ_t, the measure of vacancies is

$$v_t = \theta_t \left[u_t + s(1 - u_t) \right],$$

and the aggregate vacancy posting cost is κv_t.

The planner chooses labor market tightness θ_t, unemployment u_t, job-to-job mobility decisions, and quit decisions to maximize

$$\max_{\{\theta_t, 1_t^{EE}, 1_t^{Q}\}_{t \geq 0}} \sum_{t=0}^{\infty} \beta^t \left[\sum_{z,a,\xi} (z + a + \xi) D_t(z, a, \xi) + u_t \bar{y} - \kappa \theta_t \left(u_t + s(1 - u_t) \right) \right]. \tag{15}$$

The indicator $1_t^Q(z, a, \xi)$ equals one if a worker who draws match value (z, a, ξ) quits into unemployment. The indicator $1_t^{EE}(z_I, a_I, \xi, z_P, a_P)$ equals one if a worker currently employed at an incumbent firm with state (z_I, a_I, ξ) switches to a poaching firm with productivity z_P and amenity a_P.

The law of motion for the distribution of employed workers is

$$\begin{aligned} D_{t+1}(z, a, \xi) = {} & (1 - \delta) \left[1 - s\lambda(\theta_t) \right] D_t(z, a, \xi) \\ & + (1 - \delta) s\lambda(\theta_t) \sum_{z_P, a_P} \left[1 - 1_t^{EE}(z, a, \xi, z_P, a_P) \right] D_t(z, a, \xi) \\ & + (1 - \delta) s\lambda(\theta_t) \sum_{z_I, a_I, \xi_I} 1_t^{EE}(z_I, a_I, \xi_I, z, a) \left[1 - 1_t^{Q}(z, a, \xi) \right] D_t(z_I, a_I, \xi_I) \pi_\xi(\xi) \\ & + \lambda(\theta_t) u_t \pi_z(z) \pi_a(a) \pi_\xi(\xi) \left[1 - 1_t^{Q}(z, a, \xi) \right]. \end{aligned} \tag{16}$$

The first line captures workers whose matches survive and who do not receive outside offers. The second line captures workers who receive outside offers but remain with their incumbent employer. The third line captures workers poached from other firms who draw match-specific preference ξ at the new job and do not immediately quit. The fourth line captures unemployed workers who meet a firm, draw preference ξ, and accept the job.

In a stationary allocation, the planner chooses constant policies and a stationary distribution $D(z, a, \xi)$ satisfying (16). The stationary objective can be written as

$$\frac{1}{1-\beta}\left[\sum_{z,a,\xi}(z+a+\xi)D(z,a,\xi)+u\bar{y}-\kappa\theta\left(u+s(1-u)\right)\right], \tag{17}$$

subject to the stationary law of motion and

$$u = 1 - \sum_{z,a,\xi} D(z,a,\xi).$$

Let $S(z, a, \xi)$ denote the planner's value of assigning a worker to a match with productivity z, amenity a, and match-specific preference ξ. Let U^P denote the planner's value of unemployment. The shadow value of an additional worker in match (z, a, ξ) is the value of that match relative to unemployment:

$$\mu(z,a,\xi) = S(z,a,\xi) - U^P.$$

The planner optimality condition implies that the worker stays in match (z, a, ξ) if and only if

$$\mu(z,a,\xi) \geq 0 \quad \Leftrightarrow \quad S(z,a,\xi) \geq U^P.$$

Thus, quit and unemployment-to-employment decisions are governed by the same cutoff $\xi^Q(z, a)$ such that

$$S(z,a,\xi^Q(z,a)) = U^P.$$

Workers with $\xi < \xi^Q(z, a)$ are assigned to unemployment, while workers with $\xi \geq \xi^Q(z, a)$ are assigned to the job.

For job-to-job transitions, consider a worker currently employed in match (z, a, ξ) who receives an opportunity from a firm with productivity z' and amenity a'. If the worker moves, she draws a new match-specific preference ξ'. Define the expected social value of the new firm as

$$\widetilde{S}(z',a') = \mathbb{E}_{\xi'}\left[\max\{S(z',a',\xi'),U^P\}\right].$$

From the optimality condition, the planner reallocates the worker if and only if

$$\widetilde{S}(z',a') \geq S(z,a,\xi) \quad \Leftrightarrow \quad \mathbb{E}_{\xi'}\left[\max\{\mu(z',a',\xi'),0\}\right] \geq \mu(z,a,\xi).$$

Therefore, for a given labor market tightness θ, the planner's allocation rules are

$$\begin{aligned} 1^{Q}(z,a,\xi) &= \mathbb{1}\{S(z,a,\xi) < U^{P}\}, \\ 1^{UE}(z,a,\xi) &= \mathbb{1}\{S(z,a,\xi) \geq U^{P}\}, \\ 1^{EE}(z,a,\xi,z',a') &= \mathbb{1}\left\{\mathbb{E}_{\xi'}\left[\max\{S(z',a',\xi'),U^{P}\}\right] \geq S(z,a,\xi)\right\}. \end{aligned}$$

Thus, for a given aggregate labor market tightness, the planner allocates workers according to the present value of the match, that depends on $z+a+\xi$. However, in the decentralized economy workers respond to wages, which are not explicitly a function of preferences ξ and could reflect heterogeneous rents across jobs. We illustrate the logic behind the inefficiency with some examples.

Inefficient UE. Inefficiencies in unemployment-to-employment transitions arise because the decentralized acceptance decision differs from the planner's assignment rule. On the one hand, monopsony reduces the job finding rate: firms do not internalize the full social value of employment when posting vacancies and setting wages, so some jobs with positive surplus are not created. This channel is standard in models of monopsony: some workers with a low ξ do not work because firms offer wages below the marginal product. This force lowers the unemployment-to-employment rate relative to the constrained efficient allocation. On the other hand, conditional on receiving an offer, workers choose whether to accept based on their private value, which depends on wages rather than total match surplus. Since wages reflect heterogeneous rents across jobs, workers may accept low-surplus jobs that pay relatively high wages instead of waiting for higher-surplus jobs. This misallocation force can push unemployment-to-employment transitions above the efficient level, since workers do not internalize the benefit of searching for better jobs. Thus, the overall distortion in UE mobility is ambiguous: monopsony discourages job creation, while heterogeneous rents across jobs distort search incentives and so which jobs workers accept.

Inefficient EE. Inefficiencies in job-to-job transitions arise for the same reason: workers compare private values, while the planner compares total surplus. This creates two opposing distortions. First, workers may switch too often across similar firms. If two jobs have similar surplus but the incumbent extracts a larger rent, the worker may move to the poacher simply to obtain a higher rent even though the move does little to improve match surplus. Moreover, workers' value depends on the employer beliefs about their preferences, which drives worker mobility decision, but the social value does not depend on beliefs. As a result, workers' decision to switch jobs can be driven by a desire to prevent firms from learning about

their preferences and thus extracting higher surplus. This force raises employer-to-employer mobility relative to the efficient allocation. Second, workers may switch too little toward higher-surplus firms. High-productivity or high-amenity firms may also have larger rents, so the private wage gain from moving to them may be too small even when the social surplus gain is large. This force lowers employer-to-employer mobility. Thus, job-to-job mobility is not necessarily welfare-improving: workers may move not because the new match creates more surplus, but because the new employer extracts a smaller rent.

Constrained planner problem. Consider the problem of a constrained social planner –where the planner is subject to the same information frictions as the agents. In particular, the planner does not directly observe worker preferences. If such constrained planner could implement the relaxed planner's problem discussed above, then the constrained efficient allocation is indeed the solution to the relaxed planner's problem.

The instruments of the planner are labor taxes and subsidies contingent on the information set of firms (i.e., current wages, productivity, amenities, and beliefs about worker preferences) and a tax on vacancy posting to affect aggregate market tightness. Whenever an employed worker meets a poaching firm, the planner can impose taxes or subsidies on incumbent and poaching firms, denoted $\tau^I(w, z_I, a_I, \mathcal{B}, z_P, a_P)$ and $\tau^P(w, z_I, a_I, \mathcal{B}, z_P, a_P))$, respectively. These taxes are paid by each firm and rebated lump-sum to the same firm.

Lemma 2. *The planner can implement the allocation from the relaxed planner's problem by choosing state-contingent wage taxes* $\tau^I(w, z_I, a_I, \mathcal{B}, z_P, a_P)$, $\tau^P(w, z_I, a_I, \mathcal{B}, z_P, a_P))$, $\tau^U(z, a)$ *and a tax on vacancy posting.*

Proof: see Appendix A.

The main intuition behind this result is that the planner does not need to implement policies that depend directly on worker preferences ξ, so it can implement the solution to the relaxed planner problem without observing ξ. Indeed, the planner only wants to implement a cutoff rule $\xi^*(z_I, a_I, z_P, z_P)$ that depends on observable firms' characteristics.

5 Quantitative Results

This section quantifies the role of private worker preferences in shaping wage dynamics, worker mobility and sorting. First, we estimate the model using moments from administrative data on the labor market in France, and show that fits well several untargeted moments.

We use the model to measure implied wage markdowns and rents, with implications for inequalities. We study the inefficiencies by quantifying how far is the competitive equilibrium from the efficient allocation, with key implications for inefficient job-to-job mobility. Finally, we use the model to study the labor market effects of salary history bans, namely policies that restrict the information prospective employers can use during the hiring process.

5.1 Calibration and Model Fit

Calibration. We estimate the model by simulated method of moments. We interpret one period in the model as one quarter. We set a subset of parameters to standard values and estimate the remaining parameters to match labor market transition rates and features of job-to-job transitions. We set the discount factor to $\beta = 0.99$. The matching technology is Cobb–Douglas: we normalize matching efficiency to 0.5 and set the matching elasticity to 0.5, following Petrongolo and Pissarides (2001); Ljungqvist and Sargent (2017). The flow-utility of unemployed workers $\bar{y}$ is normalized to 0.6 as in Ljungqvist and Sargent (2017).

The remaining parameters $(\kappa, s, \delta, \sigma_z, \sigma_a, \sigma_\xi)$ are jointly estimated by targeting two groups of moments. First, we target standard labor market flows: the unemployment-to-employment rate, the employment-to-employment rate, and the employment-to-unemployment rate. These moments discipline the vacancy posting cost κ, the efficiency of on-the-job search s, and the exogenous separation rate δ. Second, we target moments that describe how productivity, wages, and firm values change during job-to-job transitions: the average change in productivity, the share of transitions involving wage cuts, and the share of transitions toward lower-ranked firms. These moments discipline the dispersion of productivity σ_z, the dispersion of firm-specific amenities σ_a, and the dispersion of match-specific preferences σ_ξ.

To see why these moments are informative, consider each source of heterogeneity in turn. Greater dispersion in firm productivity, σ_z, increases the productivity gap between randomly matched firms and therefore raises the potential productivity gains from job-to-job transitions. Greater dispersion in firm-specific amenities, σ_a, makes non-wage job characteristics more important for mobility; as a result, workers are more willing to accept wage cuts when moving to firms with better amenities. Finally, greater dispersion in match-specific preferences, σ_ξ, increases the share of job-to-job transitions toward lower-ranked firms. Without idiosyncratic preferences, workers would rarely move toward firms with lower common value. By contrast, when dispersion in idiosyncratic preferences is large, job-to-job transitions are driven more strongly by match-specific tastes. In that case, it becomes more common to observe workers moving toward lower-ranked firms, because these moves may be attractive to the worker even when the destination firm has lower common value.

Table 4: Moments and Parameters

	Data	Model	Parameter	Value
UE rate	24%	24%	Cost of posting vacancy κ	0.31
EE rate	2.4%	2.4%	On-the-job search efficiency s	0.24
EU rate	1.7%	1.7%	Exogenous separation rate δ	0.015
$E[\Delta z \mid EE]$	2.0%	2.3%	S.d. of productivity σ_z	0.10
$\Pr(\Delta w < 0 \mid EE)$	30%	30%	S.d. of amenities σ_a	0.09
$\Pr(\Delta V < 0 \mid EE)$	31%	31%	S.d. of preferences σ_ξ	0.12

Table 4 reports the targeted moments and the estimated parameters. The model matches the labor market transition rates closely. The estimated on-the-job search efficiency is $s = 0.24$, and the exogenous separation rate is $\delta = 0.015$. The estimated vacancy posting cost is $k = 0.31$. Figure 4 show the entire distribution of changes in wages and in firm desirability rank at job-to-job transitions implied by the model.[8]

The estimates imply substantial dispersion in both firm-level amenities and match-specific preferences. The standard deviation of firm productivity is $\sigma_z = 0.10$, while the standard deviation of firm-specific amenities is $\sigma_a = 0.09$ and the standard deviation of match-specific preferences is $\sigma_\xi = 0.12$. Thus, the dispersion in preferences for jobs is at least as large as the dispersion in productivity.[9] This is important because it implies that job-to-job transitions are shaped not only by movements toward more productive firms, but also by non-wage amenities and worker-specific preferences, consistent with the evidence documented in Section 2.

Validation and model fit. We next evaluate the model's ability to match key untargeted moments of interest. First, we compare the average wage markdown implied by the model to existing estimates in the literature. The model implies an average wage markdown of approximately 20%, measured as the gap between wages and marginal products. This magnitude is close to existing estimates of markdowns in the labor market (Berger, Herkenhoff, and Mongey, 2022; Lamadon, Mogstad, and Setzler, 2022). Second, we compare the model-implied acceptance rate of job offers from unemployment to empirical estimates. In the model, dispersion in unobserved idiosyncratic preferences implies that unemployed workers do not accept all job offers: even when a firm makes an offer, the worker may draw a low preference for the job and reject it. The model implies an acceptance rate from unemployment of approximately 50%, in line with the evidence in Faberman et al. (2022).

[8] Change in desirability rank is an ordinal measure. For instance, a change of 0.4 means that the destination firm is approximately 1.4 more desirable than the initial firm.

[9] Note that the flow value of a match is additive in its elements z, a, ξ, so these standard deviations are expressed in the same units.

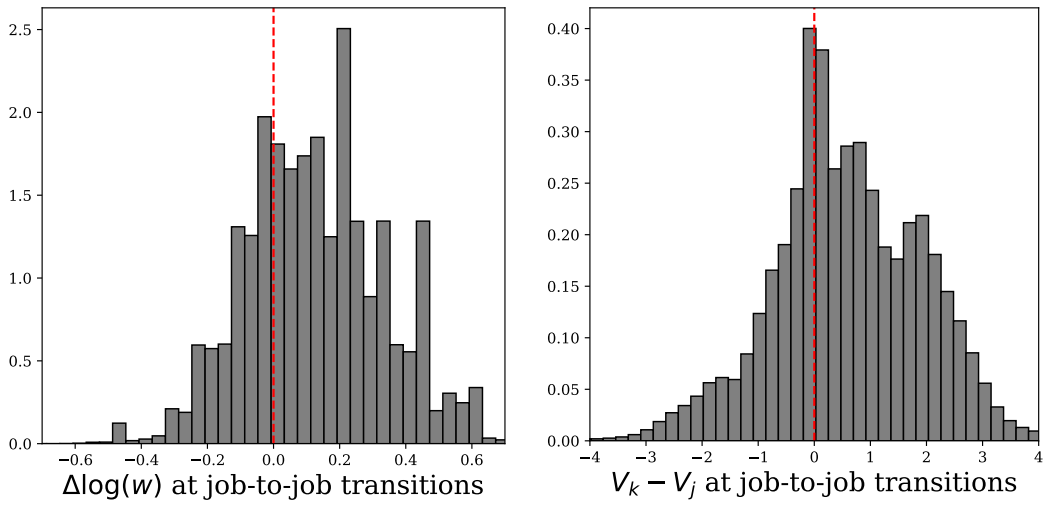


Figure 4: Changes in wages and in firm rankings at job-to-job transitions

Third, we compare the size of worker preferences for jobs implied by the model to survey evidence. Caldwell, Haegele, and Heining (2025) use survey questions to elicit the wage increase workers would require to move to an otherwise identical firm. This comparison is designed to hold fixed firm-level amenities and distance to work, so the elicited premium captures workers' preference for their current employer. They find that the required wage premium lies between 7% and 18% of the current wage. In our model, the wage premium required to move to an otherwise identical job with the same (z, a) is approximately 10% of the current wage. This premium arises because workers sort into jobs they like: although the idiosyncratic preference ξ is drawn at random when a worker joins a firm, employed workers in equilibrium have higher preferences for their current jobs than the average draw in the population.

Fourth, we show that the model replicates the tenure profile of job-to-job transitions toward lower-ranked firms. As documented in Section 2, workers with higher tenure are less likely to move toward lower-ranked firms. The model is consistent with this pattern, as shown in Figure 5. The mechanism is the learning channel described in Section 4. Workers may have incentives to leave jobs they like in order to avoid revealing their attachment to the incumbent firm, since revealing this information would weaken their future bargaining position. These incentives become stronger with tenure, as firms accumulate more information about workers' preferences. As a result, high-tenure workers who do move are more likely to move toward higher-ranked firms. To isolate the role of learning, we solve a counterfactual economy in which firms behave naively and do not update beliefs about workers' preferences. In that economy, the share of job-to-job transitions toward lower-ranked firms varies little with

tenure, precisely because workers no longer have an incentive to leave jobs they like in order to prevent firms from learning about their preferences.

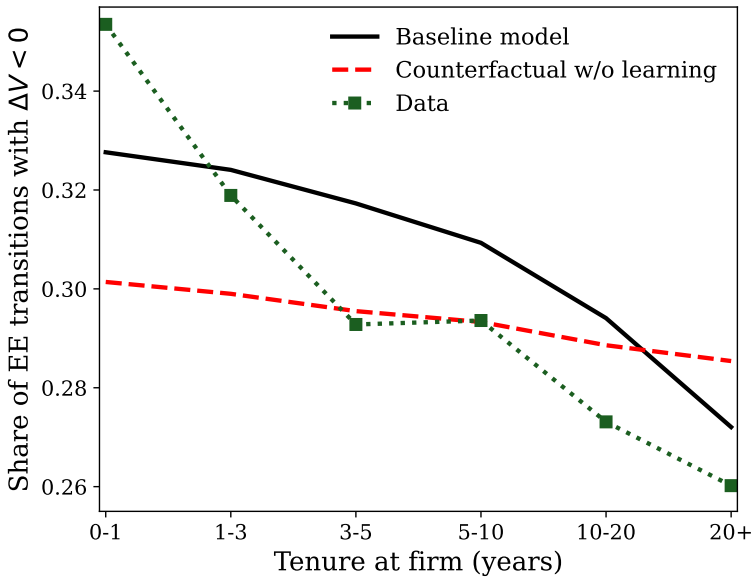


Figure 5: Job-to-job transitions toward lower ranked firms and workers' tenure.

5.2 Markdowns, Rents, and Inequality

The estimated model implies an average wage markdown of approximately 20%, measured as the average gap between the wage and the marginal product of labor. Several forces contribute to this markdown. First, markdowns reflect compensating differentials associated with observable non-wage amenities and, potentially, with worker-specific preferences that firms learn about over time. Second, markdowns arise because firms face upward-sloping labor supply curves due to unobserved worker preferences. Third, search frictions limit workers' outside options and therefore allow firms to pay wages below marginal products.

Markdowns are heterogeneous both across and within firm types (z, a). Across firm types, markdowns vary because firms differ in productivity and amenities. Firms with different (z, a) may face different labor supply elasticities and different outside-option environments. For example, high-productivity firms may face less competition from even more productive firms, which affects the outside offers available to their workers. In addition, firms with more valuable amenities may pay lower wages as compensating differentials. Within a firm type, markdowns vary because workers differ in their labor market histories. Workers employed at the same type of firm may have received different outside offers over time, and firms may have formed different beliefs about their idiosyncratic preferences for the job.

Quantitatively, we find that about 40% of the dispersion in wage markdowns is explained by variation across firm types, while the remaining 60% arises within firm types. This distinction matters for wage inequality. If high-productivity firms charge larger markdowns, markdowns can reduce wage inequality by compressing the wages of workers who would otherwise be paid more. To quantify these forces, we decompose the variance of wages into the variance of marginal products, the variance of markdowns, and their covariance. We find that productivity differences account for 44% of wage dispersion, while heterogeneity in markdowns accounts for 57%. The remaining component is explained by the covariance between marginal products and markdowns.

Wage markdowns, however, are not sufficient to draw welfare conclusions about inequality. Workers derive utility not only from wages but also from firm-level amenities and idiosyncratic preferences for jobs. Their flow utility is $(w + a + \xi)$. For this reason, a more relevant measure of inequality is the dispersion of workers' flow utility rather than the dispersion of wages alone. We decompose the variance of $w + a + \xi$ into the variance of wages, the variance of firm amenities, the variance of idiosyncratic preferences, and the associated covariance terms.

This decomposition shows that wage dispersion accounts for approximately 80% of inequality in flow utility. Firm-specific amenities are also dispersed, but their contribution is almost entirely offset by covariance terms. In contrast, idiosyncratic preferences account for approximately 20% of inequality, because the covariance terms involving ξ do not fully offset its variance. This result highlights two implications of our framework. First, whether non-wage amenities are observed by employers matters for inequality and for the effects of redistributive policies.[10] Second, even though employers can learn about workers' idiosyncratic preferences over time, heterogeneity in these preferences remains quantitatively important for inequality because wages only partially offset the utility workers derive from their preferred jobs.

5.3 Inefficient Worker Mobility

We now quantify the aggregate implications of inefficiency due to wage setting, worker mobility, and job creation. Section 4.3 presented the problem of a social planner and it shows that the decentralized equilibrium is not constrained efficient. Inefficiencies are driven by two channels. First, workers make mobility decisions based on private values, while the planner

[10]For instance, suppose high-wage workers are compensated for low non-wage amenities. In that case, a progressive income tax could increase inequality in utility rather than reduce it, because it taxes workers who already receive low non-wage compensation. By contrast, if high-wage workers also enjoy high utility from unobserved idiosyncratic preferences, a progressive income tax would reduce inequality more substantially.

allocates workers based on total match surplus. Second, equilibrium wages might lead to too little or too much job creation through vacancy posting. In this section, we compare the decentralized equilibrium to the constrained efficient allocation. This comparison allows us to quantify how much worker mobility, sorting, and welfare are distorted by equilibrium wage setting.

The key mechanism driving inefficient worker mobility is that workers internalize

$$w + a + \xi,$$

while the planner internalizes

$$z + a + \xi.$$

Thus, workers respond to wages, which reflect heterogeneous markdowns and rents, whereas the planner responds to total match surplus. If workers' share of surplus was constant across jobs, then workers would rank jobs in the same way as the planner. This is often the case in models with full information in which the workers have a constant bargaining share (Cahuc, Postel-Vinay, and Robin, 2006; Berger et al., 2023; Lamadon et al., 2024). Inefficiencies arise because worker rents differ across firms and workers, and wages are not contingent on workers' preferences. As a result, workers may move toward jobs that entitle them to larger rents and not toward jobs with higher surplus.

The theory in Section 4.3 highlights two opposing forces. First, workers may switch too often across similar firms. When two jobs generate similar surplus, a worker may leave the incumbent simply because the poaching firm extracts a smaller rent relative to an incumbent who acquired information about the worker's preference for the current job. In addition, leaving can prevent the incumbent from learning that the worker strongly values her current job, which would weaken her future bargaining position if she stayed with the incumbent. This force raises employer-to-employer mobility relative to the efficient allocation. Second, workers may switch too little toward higher-surplus firms. High-productivity or high-amenity firms may also charge larger markdowns, so the private wage gain from moving to these firms may be too small even when the social surplus gain is large. This force lowers reallocation toward high-surplus jobs. A similar tension arises for worker decision to accept jobs from unemployment, where the job finding rate might be either too high or too low in the competitive equilibrium.

Table 5 compares the decentralized equilibrium to the planner allocation. Welfare is about 2.2% lower in the decentralized equilibrium relative to the full constrained efficient allocation. The equilibrium features substantially more job-to-job mobility: the employer-to-employer transition rate is 2.4%, compared with 1.6% in the full planner allocation. Hence,

at least one third of job-to-job transitions in the decentralized equilibrium are inefficient. These transitions are privately optimal for workers, but they do not increase total match surplus.

The table also reports a planner allocation that holds labor market tightness fixed at its equilibrium level. This comparison isolates the partial-equilibrium mobility distortion from the general-equilibrium vacancy-creation margin. Holding θ fixed, the planner reduces the employer-to-employer transition rate from 2.4% to 1.9%. Thus, even fixing aggregate labor market conditions, the decentralized economy features too much employer-to-employer mobility. The full planner further reduces the EE rate to 1.6%, reflecting both more efficient mobility decisions and the general-equilibrium adjustment in vacancy creation. Quantitatively, however, we find that a substantial share of the welfare losses are driven by the inefficiencies in general equilibrium. Indeed, while workers switch jobs too often in the competitive equilibrium, the implied welfare looses keeping constant θ are modest and equal to 0.1%. Intuitively, this is because workers leave disproportionately more jobs that they like to sort into jobs with higher productivity, leading to lower average preference ξ and higher average productivity z in the competitive equilibrium.

A key implication of the quantitative analysis is that job-to-job mobility is not necessarily welfare enhancing, and many workers move across employers not because the new job generates higher surplus, but because it offers the worker a higher rent.

Table 5: Aggregate Effects of Inefficient Worker Mobility

	Equilibrium	Planner (Fixed θ at Eq.)	Planner (GE)	Difference (Equilibrium−Planner)
Total welfare	110.7	110.8	113.2	−2.2%
Unemployment rate	6.4%	6.5%	8.7%	−2.2pp
EE transition rate	2.4%	1.9%	1.6%	+0.8pp
Average productivity z	1.11	1.10	1.09	+2pp
Average amenity	0.07	0.08	0.07	0pp
Average preference ξ	0.07	0.08	0.075	−0.5pp

5.4 The Labor Market Effects of Salary History Bans

There has been growing interest in policies that restrict the information prospective employers can use during the hiring process. A prominent example is a *salary history ban*, which prevents employers from asking job applicants about their current or previous wages. For instance, the 2023 EU Pay Transparency Directive prevents employers from asking applicants

about their pay history. Similar policies have been adopted in several U.S. jurisdictions. New York City made it illegal for employers to ask about salary history during the hiring process in 2017, while Massachusetts and California introduced related restrictions as part of broader equal-pay legislation.

The intended goal of these policies is to reduce inequality and weaken the link between past and future wages: if a worker is currently underpaid, the next employer should be less able to use this information to offer a low wage and thereby perpetuate the initial wage disadvantage. At the same time, opponents argued that limiting the information available to prospective employers could have unintended consequences.[11] Yet, existing work has not provided a quantitative assessment of the equilibrium effects of these policies, and empirical studies find mixed results (Agan, Cowgill, and Gee, 2021; Khanna, 2020; Tunc, 2025).

We use our model to study the labor-market effects of these policies. The model provides a natural laboratory for two reasons. First, in standard full-information labor market models, past salaries do not directly affect wage setting. In our model, by contrast, firms compete for workers in an environment with incomplete and asymmetric information, and they value information about workers' labor-market histories. This makes the model well suited to study policies that restrict the information available to prospective employers about workers' labor market history. Second, firms in the model would like to implement a form of price discrimination across workers. In particular, a worker who earns a low wage in the current job is more likely to receive a low wage offer from a prospective employer. Importantly, prospective employers can implement this form of price discrimination because they observe labor-market histories of prospective employees, even though workers' preferences are unobserved. Thus, the ability of firms to infer worker preferences from observables is central to wage setting in the model.

We interpret a salary history ban as a policy that impairs the ability of prospective employers to observe workers' labor-market histories, including their current wage. In the counterfactual economy, the only information available to a prospective employer is the identity of the worker's current employer, if the worker is employed. This assumption captures the idea that the identity of the current employer is difficult to conceal, since it can typically be inferred from résumés, professional networks, or social media platforms. We also assume workers cannot volunteer information about their current wage or about their labor market experience.[12] Before turning to the quantitative effects of the policy, we describe how it

[11] For instance, the Greater Philadelphia Chamber of Commerce filed a lawsuit challenging the validity of the law and moved for a preliminary injunction. More importantly, two states, Michigan and Wisconsin, passed laws explicitly prohibiting local governments within the state from adopting such policies, with the intention of protecting employers' right to solicit salary information from prospective employees.

[12] This assumption captures the best-case scenario for the policy, in which pay-history bans fully remove

changes the decision problem of workers, incumbent firms, and poaching firms. The policy mainly affects the decision problem of poaching firms.

We assume that firm types are always observed by both workers and firms. Thus, the policy limits the ability of poaching firms to acquire information about the worker's pay history, but it does not limit information about the identity or type of the incumbent and prospective employers. Consider first the employed worker's problem. The worker's decision problem is unchanged: the worker takes as given the wages offered by the incumbent and poaching firms and optimally chooses whether to stay or move. The policy also does not affect the amount of information revealed by the worker's decision. If the worker stays with the incumbent, the incumbent has the same information set as in the baseline model. Moreover, we assume that the incumbent observes the wage offered by the poaching firm.[13] Thus, conditional on the offered wages, the worker's decision to stay is equally informative to the incumbent firm as in the baseline economy.

The incumbent firm's problem is also unchanged conditional on the wage offered by the poaching firm. The incumbent chooses its wage to maximize the same objective as in the baseline model, using the same information set and the same law of motion for beliefs. The key difference concerns the poaching firm. In the baseline model, the poacher observes the worker's current wage w and the incumbent's belief $\mathcal{B}$, which summarizes the information that the incumbent has accumulated about the worker's preferences. Under the salary history ban, the poacher does not observe either w or $\mathcal{B}$. Instead, it chooses a wage taking as given the incumbent's wage policy and forming beliefs about the unobserved states. In a stationary Bayesian-Nash equilibrium, these beliefs coincide with the stationary distribution of $(w, \mathcal{B})$ conditional on the types of the incumbent and poaching firms.

Formally, consider a worker employed at an incumbent firm with type (z_I, a_I) who meets a poaching firm with type (z_P, a_P). Under the salary history ban, the poacher observes the firm types but does not observe the worker's current wage w or the incumbent's belief $\mathcal{B}$ about the worker's preference for the current job. The poacher therefore chooses a wage offer $w_P(z_I, a_I, z_P, a_P)$ that maximizes expected profits over the distribution of unobserved

information about current and past wages from the poacher's information set. In our model, current wages and past labor market choices affect wage offers because they shape firms' beliefs about workers' preferences. It is therefore unclear whether workers could *credibly* disclose information about their current wage but also about their past actions.

[13] An alternative exercise would consider a counterfactual economy in which the incumbent firm does not observe the wage offered by the poaching firm. In the counterfactual studied here, however, the poaching wage depends only on the types of the incumbent and poaching firms. Therefore, knowing the type of the poaching firm is sufficient for the incumbent to infer the wage offered by the poacher.

states, given the wage schedule of the incumbent:

$$w_P \in \arg\max_{w_P} \int \Pi^P \left(w_P, w_I(\mathcal{B}, w), z_I, a_I, z_P, a_P, \mathcal{B}, w\right) \, d\mu(w, \mathcal{B} \mid z_I, a_I, z_P, a_P). \tag{18}$$

Here, $w_I(w, z_I, a_I, z_P, a_P, \mathcal{B})$ is the incumbent wage policy, $\mu(w, \mathcal{B} \mid z_I, a_I, z_P, a_P)$ is the poacher's equilibrium belief over the worker's unobserved current wage and the incumbent's belief, and $\Pi^P(\cdot)$ denotes the poacher's expected profit conditional on the worker's state. Thus, under the salary history ban, the poaching firm cannot condition its offer on the worker's current wage or on the incumbent's belief. Instead, it chooses a wage by averaging profits across all possible unobserved states $(w, \mathcal{B})$, weighting each state by its conditional stationary probability.

Before presenting the quantitative results, it is useful to discuss the mechanisms through which the policy affects labor-market outcomes. Start with wages. In the baseline equilibrium, the wage offered by the poaching firm is increasing in the incumbent's belief $\mathcal{B}$. When the poacher no longer observes $\mathcal{B}$, it can no longer tailor wage offers to this belief. This has two opposing effects. On the one hand, workers with low $\mathcal{B}$ may receive higher wage offers, because the poaching firm can no longer identify them as workers who are relatively attached to their current job. These workers benefit from an informational rent: their wage increases because the prospective employer does not know that they have a low willingness to move. On the other hand, workers with high $\mathcal{B}$ may receive lower wage offers, because the poaching firm can no longer target them with especially high offers. These workers face weaker competition, since the poaching firm has less information with which to tailor its wage offer.

A similar logic applies to the current wage. In the baseline equilibrium, the poaching wage is non-decreasing in the worker's current wage at the incumbent firm. When the poacher does not observe the current wage, some low-wage workers receive higher offers because the poacher can no longer condition on their low current pay. At the same time, some high-wage workers receive lower offers because the poacher no longer observes that a high offer may be necessary to attract them. In equilibrium, these effects also spill over to incumbent wage setting and to wages offered to unemployed workers, because wages are strategic complements and because the policy changes workers' outside options.

The policy also affects the allocation of workers across firms. By creating additional information frictions, a salary history ban reduces the ability of poaching firms to tailor wage offers to worker-specific information. This can worsen the allocation of workers across jobs. For example, consider a worker who strongly prefers the current job and whose incumbent firm has learned this preference over time. In the baseline economy, a poaching firm that

observes this belief may offer a high wage when the new match is sufficiently productive. Under the salary history ban, the poacher cannot condition its offer on the incumbent's belief and may therefore offer a lower wage. As a result, the poacher may fail to attract the worker even when the new match would generate higher surplus than the existing match. Thus, while the policy reduces firms' ability to price discriminate against workers, it may also reduce the efficiency of worker reallocation.

Table 6 compares the steady state of the baseline economy to the steady state of the counterfactual economy in which poaching firms do not observe workers' labor-market histories. We find that salary history bans increase average wages by approximately 6 percent. We also find that salary history bans reduce wage inequality, since wage offers become more homogeneous across workers. The standard deviation of wages falls from 17.% of the average wage to 13.5% of the average wage. Similarly, the gap between the 90th and the 10th percentile falls from 45% of the average wage to 34.4% of the average wage. This does not imply that every worker gains: some workers receive lower wages than in the baseline economy because poaching firms can no longer target them with high offers. However, quantitatively, most workers experience higher wages, and even the 95th percentile of the wage distribution is higher under the salary history ban than in the baseline economy. This result reflects the strategic complementarity in wage setting: higher poaching wages induce higher incumbent wages, which in turn raise workers' outside options more broadly. Salary history bans also reduce unemployment: higher average wages increase the value of employment relative to unemployment, making workers more willing to accept job offers.

However, the increase in wages comes with an efficiency cost. By weakening the ability of firms to condition wage offers on worker-specific information, the policy worsens the allocation of workers across matches. As a result, the average match productivity and match value decline by 1.35% and 0.72% respectively. Thus, the counterfactual highlights a central trade-off: salary history bans raise wages and reduce wage inequality by limiting firms' ability to price discriminate, but they also introduce additional information frictions that can reduce allocative efficiency. Despite the average match value falls, we find that total welfare increases with salary history bans. This result is driven by two forces. First, lower unemployment rate partially offset the implied decline in welfare from lower match value. Second, salary history bank shrink firms' profits and vacancy creation, which leads to higher welfare because the competitive equilibrium features too much vacancy posting.

Table 6: Effects of Salary History Bans

	(1) Baseline	(2) Salary History Ban	(3) Difference: (2) - (1)
Average wage	0.908	0.962	+5.9%
Wage inequality: $sd(w)/E[w]$	17.3%	13.5%	-3.8pp
Wage inequality: $(p90 - p10)/E[w]$	45.1%	34.4%	-10.7pp
Average productivity	1.114	1.099	-1.35%
Average match value	1.259	1.250	-0.72%
Unemployment rate	6.48%	5.66%	-0.82pp
EE transition rate	2.44%	2.48%	+0.04pp
Job finding rate	24.27%	26.53%	+2.26pp
$\Pr(\Delta V < 0 \mid EE)$	32.97%	37.00%	+4.03pp
Welfare	110.62	112.12	+1.35%

6 Conclusion

This paper develops a dynamic search model in which workers have heterogeneous preferences for jobs that are private information. Because firms do not observe how much workers value their jobs, they set wages based on beliefs, and these beliefs evolve endogenously through workers' mobility decisions. We estimate the model using French matched employer-employee data and show that private preferences are quantitatively important for wages, markdowns, and worker mobility. The model generates substantial markdown heterogeneity both across and within firms, and this heterogeneity distorts labor market flows: workers sometimes move not toward jobs with higher surplus, but toward jobs where they can capture larger rents. As a result, at least one third of job-to-job transitions are inefficient. The model also shows that policies limiting information available to prospective employers, such as salary history bans, can raise wages and reduce inequality, but at the cost of lower allocative efficiency. More broadly, the results highlight that worker preferences for jobs are not only a source of monopsony power, but also a key determinant of wage dynamics, sorting, and allocative efficiency.

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A Model Appendix and Proofs

A.1 Proof of Lemma 1

The reservation threshold

$$\xi^* = \xi^*(w_I, w_P, z_I, a_I, z_P, a_P, \mathcal{B})$$

is implicitly defined by

$$V\left(w_I, z_I, a_I, \xi^*, \mathcal{B}'(w_I, w_P)\right) = \mathbb{E}_{\xi_P}\left[\max\left\{V\left(w_P, z_P, a_P, \xi_P, \mathcal{B}'_U(w_P)\right), U\right\}\right]. \tag{19}$$

Consider first an increase in the incumbent wage w_I, holding $(w_P, z_I, a_I, z_P, a_P, \mathcal{B})$ fixed. The worker's value from staying with the incumbent is increasing in the wage. Therefore, for any given cutoff ξ^*, an increase in w_I raises the left-hand side of (19), while the right-hand side is unchanged. To restore equality, the cutoff ξ^* must decrease. Hence, ξ^* is decreasing in w_I.

Now consider an increase in the poaching wage w_P, holding $(w_I, z_I, a_I, z_P, a_P, \mathcal{B})$ fixed. The worker's value from switching to the poacher is increasing in w_P. Therefore, for any given cutoff ξ^*, an increase in w_P raises the right-hand side of (19), while the left-hand side is unchanged. To restore equality, the cutoff ξ^* must increase. Hence, ξ^* is increasing in w_P.

This establishes the result. □

A.2 Proof of Proposition 1

The next two proofs assume that the distribution of preferences is Uniform. Under this assumption, beliefs can be summarized by the lower support of the posterior distribution, denoted by $\underline{\mathcal{B}}$. The incumbent chooses w_I to maximize

$$G_I(w_I; w_P, z_I, a_I, z_P, a_P, \underline{\mathcal{B}}) =$$
$$\left[1 - F\left(\xi^*(w_I, w_P, z_I, a_I, z_P, a_P, \underline{\mathcal{B}}) \mid \underline{\mathcal{B}}\right)\right] \Pi\left(w_I, z_I, a_I, \xi^*(w_I, w_P, z_I, a_I, z_P, a_P, \underline{\mathcal{B}})\right),$$

where, with a slight abuse of notation, $\Pi(w_I, z_I, a_I, \xi^*)$ denotes the value of the filled job when the posterior belief is summarized by lower support ξ^*.

We distinguish two cases.

Case 1: $\xi^* > \underline{\mathcal{B}}$**.** Under the Uniform distribution,

$$1 - F(\xi^* \mid \underline{\mathcal{B}}) = \frac{\bar{\xi} - \xi^*}{\bar{\xi} - \underline{\mathcal{B}}},$$

where $\bar{\xi}$ denotes the upper bound of the distribution. Maximizing G_I is equivalent to maximizing $\log G_I$:

$$\log G_I = \log\left(\frac{\bar{\xi} - \xi^*}{\bar{\xi} - \underline{\mathcal{B}}}\right) + \log \Pi\left(w_I, z_I, a_I, \xi^*\right).$$

Conditional on wages and firm types, ξ^* does not vary mechanically with $\underline{\mathcal{B}}$ in this region. Hence,

$$\frac{\partial \log G_I}{\partial \underline{\mathcal{B}}} = \frac{1}{\bar{\xi} - \underline{\mathcal{B}}},$$

and therefore

$$\frac{\partial^2 \log G_I}{\partial w_I \, \partial \underline{\mathcal{B}}} = 0.$$

Thus, in this region the incumbent's objective has weakly increasing differences in both $(w_I, \underline{\mathcal{B}})$ and $(w_I, -\underline{\mathcal{B}})$. By Topkis' Theorem, the incumbent's best response is independent of $\underline{\mathcal{B}}$ in this region.

Case 2: $\xi^* \leq \underline{\mathcal{B}}$**.** If $\xi^* \leq \underline{\mathcal{B}}$, the worker stays with probability one under the posterior belief, since $\xi \geq \underline{\mathcal{B}}$ almost surely. In this case, the incumbent can lower the wage until the constraint $\xi^* = \underline{\mathcal{B}}$ binds. When $\underline{\mathcal{B}}$ increases, maintaining $\xi^* = \underline{\mathcal{B}}$ requires lowering w_I, since ξ^* is decreasing in w_I by Lemma 1. Hence, the incumbent's best response is decreasing in $\underline{\mathcal{B}}$ in this region.

Combining the two cases implies that the incumbent best response

$$w_I^*(w_P; z_I, a_I, z_P, a_P, \underline{\mathcal{B}})$$

is weakly decreasing in $\underline{\mathcal{B}}$. □

A.3 Proof of Proposition 2

The poaching firm chooses w_P to maximize

$$G_P(w_P; w_I, z_I, a_I, z_P, a_P, \underline{\mathcal{B}}) =$$
$$[1 - F(\xi_U^*(w_P, z_P, a_P))] F\left(\xi^*(w_I, w_P, z_I, a_I, z_P, a_P, \underline{\mathcal{B}}) \mid \underline{\mathcal{B}}\right) \Pi\left(w_P, z_P, a_P, \underline{\mathcal{B}}'_U(w_P)\right).$$

We again distinguish two cases.

Case 1: $\xi^* > \underline{\mathcal{B}}$. Under the Uniform distribution,

$$F(\xi^* \mid \underline{\mathcal{B}}) = \frac{\xi^* - \underline{\mathcal{B}}}{\bar{\xi} - \underline{\mathcal{B}}}.$$

Consider the log objective:

$$\log G_P = \log\left(\frac{\xi^* - \underline{\mathcal{B}}}{\bar{\xi} - \underline{\mathcal{B}}}\right) + \log \Pi\left(w_P, z_P, a_P, \underline{\mathcal{B}}'_U(w_P)\right).$$

The direct effect of $\underline{\mathcal{B}}$ on the first term is

$$\frac{\partial \log G_P}{\partial \underline{\mathcal{B}}} = -\frac{1}{\xi^* - \underline{\mathcal{B}}} + \frac{1}{\bar{\xi} - \underline{\mathcal{B}}} < 0.$$

Using Lemma 1, ξ^* is increasing in w_P, so $\partial \xi^* / \partial w_P > 0$. Therefore,

$$\frac{\partial^2 \log G_P}{\partial w_P \, \partial \underline{\mathcal{B}}} = \frac{1}{(\xi^* - \underline{\mathcal{B}})^2} \frac{\partial \xi^*}{\partial w_P} > 0.$$

Thus, $\log G_P$ has increasing differences in $(w_P, \underline{\mathcal{B}})$. By Topkis' Theorem, the poaching firm's best response is increasing in $\underline{\mathcal{B}}$ in this region.

Case 2: $\xi^* \leq \underline{\mathcal{B}}$. If $\xi^* \leq \underline{\mathcal{B}}$, the worker stays with probability one and the poacher attracts the worker with probability zero. Hence $G_P = 0$ and the poacher's payoff is flat in w_P. In this region, the poacher's best response is independent of $\underline{\mathcal{B}}$.

Combining the two cases implies that the poaching firm's best response

$$w_P^*(w_I; z_I, a_I, z_P, a_P, \underline{\mathcal{B}})$$

is weakly increasing in $\underline{\mathcal{B}}$. □

A.4 Proof of Proposition 3

This result relies on Proposition 1 and Proposition 2. From Proposition 1 and Proposition 2, we know that the equilibrium response of w_P to an increase in $\underline{\mathcal{B}}$ depends on

(i) the effect of a change in $\underline{\mathcal{B}}$ on w_P holding w_I fixed,

(ii) the effect of a change in $\underline{\mathcal{B}}$ on w_I holding w_P fixed,

(iii) the effect of a change in w_P on w_I,

(iv) the effect of a change in w_I on w_P.

We use Proposition 1 to argue that the effect of a change in $\underline{\mathcal{B}}$ on w_I holding w_P fixed is weakly negative. From Proposition 2, we know that the effect of a change in $\underline{\mathcal{B}}$ on w_P holding w_I fixed is weakly positive. There are then four cases, depending on the effect of w_P on the incumbent best response w_I and the effect of w_I on the poacher best response w_P. Throughout, we hold fixed firm types (z_I, a_I, z_P, a_P).

- **Case 1:** $\partial w_I/\partial w_P > 0$, $\partial w_P/\partial w_I > 0$. Then, an increase in $\underline{\mathcal{B}}$ holding w_I fixed leads to a higher w_P by Proposition 2. Then, w_I increases because $\partial w_I/\partial w_P > 0$, and w_P further increases because $\partial w_P/\partial w_I > 0$. Thus, w_P must increase.

- **Case 2:** $\partial w_I/\partial w_P < 0$, $\partial w_P/\partial w_I < 0$. Then, an increase in $\underline{\mathcal{B}}$ holding w_I fixed leads to a higher w_P by Proposition 2. Then, w_I decreases because $\partial w_I/\partial w_P < 0$, and w_P further increases because $\partial w_P/\partial w_I < 0$. Thus, w_P must increase.

- **Case 3:** $\partial w_I/\partial w_P < 0$, $\partial w_P/\partial w_I > 0$. Suppose, by contradiction, that an increase in $\underline{\mathcal{B}}$ leads to a lower w_P. Then, w_I increases because $\partial w_I/\partial w_P < 0$. However, since $\partial w_P/\partial w_I > 0$ and Proposition 2 implies a direct positive effect of $\underline{\mathcal{B}}$ on w_P, we must have that w_P increases, a contradiction.

- **Case 4:** $\partial w_I/\partial w_P > 0$, $\partial w_P/\partial w_I < 0$. Suppose, by contradiction, that an increase in $\underline{\mathcal{B}}$ leads to a lower w_P. Then, w_I decreases because $\partial w_I/\partial w_P > 0$. However, since $\partial w_P/\partial w_I < 0$ and Proposition 2 implies a direct positive effect of $\underline{\mathcal{B}}$ on w_P, we must have that w_P increases, a contradiction.

Therefore, in all cases, an increase in $\underline{\mathcal{B}}$ weakly increases the equilibrium wage offered by the poacher, w_P^*. This establishes the result. □

A.5 Proof of Lemma 2

First, we show that the constrained planner can implement the same cutoff rules as in the relaxed planner problem for: (i) the worker's decision to accept an outside offer, and (ii) the worker's decision to quit the new job at the poaching firm. We then show that the constrained planner can implement the same cutoff rule for workers who receive job offers from unemployment. Finally, the planner can use taxes and subsidies on vacancy posting to implement the optimal labor market tightness θ, given the efficient cutoff rules.

Step 1: EE. Fix a meeting between an employed worker at an incumbent firm and a poaching firm. Firm characteristics are summarized by (z_I, a_I, z_P, a_P).

Let $\xi^{PL}(z_I, a_I, z_P, a_P)$ denote the cutoff for efficient job-to-job mobility in the relaxed planner's problem.

For any pair of wage offers (w_I, w_P), the worker's switching decision in the competitive equilibrium is characterized by a cutoff ξ^* solving

$$V(w_I, z_I, a_I, \xi^*, \xi^*) = \mathbb{E}_{\xi_P}\left[V(w_P, z_P, a_P, \xi_P, \mathcal{B}'_U(w_P))\right].$$

By Lemma 1, the cutoff ξ^* is decreasing in w_I and increasing in w_P. Thus, a planner who can control wages can also control the cutoff rule.

Similarly, the worker's decision to quit after an employer-to-employer transition is characterized by the cutoff ξ^*_U such that

$$V(w_P, z_P, a_P, \xi^*_U, \xi^*_U) = U.$$

Thus, the planner can set state-contingent labor taxes or subsidies,

$$\tau^I(w_I, z_I, a_I, z_P, a_P, \mathcal{B}) \quad \text{and} \quad \tau^P(w_P, z_I, a_I, z_P, a_P, \mathcal{B}),$$

so that the implied wages w_I^{PL} and w_P^{PL} implement the cutoff rules of the relaxed planner problem:

$$V(w_I^{PL}, z_I, a_I, \xi^{PL}, \xi^{PL}) = \mathbb{E}_{\xi_P}\left[V(w_P^{PL}, z_P, a_P, \xi_P, \mathcal{B}'_U(w_P^{PL}))\right], \tag{20}$$

$$V(w_P^{PL}, z_P, a_P, \xi_U^{PL}, \xi_U^{PL}) = U. \tag{21}$$

The planner has two instruments and can therefore implement the two cutoff rules: the job-to-job switching cutoff and the quit cutoff at the poaching firm.

Step 2: UE. The worker's decision to accept a job from unemployment, or equivalently not to quit immediately after accepting it, is characterized by the cutoff ξ^*_U such that

$$V(w_U, z, a, \xi^*_U, \xi^*_U) = U.$$

Similarly, the planner can impose state-contingent wage taxes or subsidies for workers hired from unemployment to implement a wage w_U^{PL} such that

$$V(w_U^{PL}, z, a, \xi_U^{PL}, \xi_U^{PL}) = U.$$